

SchrotbergerMilestone5

August 8, 2026

1 Milestone 2

1.1 Cleaning/Formatting Flat File Source

```
[3]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
[5]: players = pd.read_csv("G:/My Drive/College/Caitie/DSC540-T301/
↳volleybox_women_players.csv")
```

Perform at least 5 data transformation and/or cleansing steps to your flat file data.

```
[7]: players.head()
```

```
[7]:
```

	Id	Name	Nickname	Maiden name	Birthdate	Nationality	\
0	p178623	Giada Maria Falsone	NaN	NaN	NaN	Italy	
1	p178618	Giorgia Polimeni	NaN	NaN	NaN	Italy	
2	p178616	Silvia Simone	NaN	NaN	NaN	Italy	
3	p178614	Martina Mazza	NaN	NaN	NaN	Italy	
4	p178613	Sophia Letizia Foti	NaN	NaN	NaN	Italy	

	Position	Dominant hand	Height (cm)	Weight (kg)	...	Followers	\
0	Universal	NaN	NaN	NaN	...	NaN	
1	Universal	NaN	NaN	NaN	...	NaN	
2	Universal	NaN	NaN	NaN	...	NaN	
3	Universal	NaN	NaN	NaN	...	NaN	
4	Universal	NaN	NaN	NaN	...	NaN	

	Player ranking	Coach ranking	Registered	Views	Last activity	\
0	NaN	NaN	No	6	NaN	
1	NaN	NaN	No	6	NaN	
2	NaN	NaN	No	6	NaN	
3	NaN	NaN	No	6	NaN	
4	NaN	NaN	No	6	NaN	

	Added by	User Recommendations	Partners
0	Mattia Mario Guido Caputo	NaN	NaN

```

1  Mattia Mario Guido Caputo  NaN          NaN      NaN
2  Mattia Mario Guido Caputo  NaN          NaN      NaN
3  Mattia Mario Guido Caputo  NaN          NaN      NaN
4  Mattia Mario Guido Caputo  NaN          NaN      NaN

```

[5 rows x 50 columns]

```
[9]: orig_len = len(players)
     orig_len
```

[9]: 78693

1.1.1 First Transformation

```
[12]: # Identify any duplicates for key or name
print("ID is duplicated: {}".format(any(players.Id.duplicated())))
print("Name is duplicated: {}".format(any(players.Name.duplicated())))
```

ID is duplicated: False

Name is duplicated: True

```
[14]: # Remove duplicated players based on name and birthday
player_dedup = players.drop_duplicates(subset=['Name', 'Birthdate'], keep='first')
player_dedup.head()
```

```
[14]:
```

	Id	Name	Nickname	Maiden name	Birthdate	Nationality	\
0	p178623	Giada Maria Falsone	NaN	NaN	NaN	Italy	
1	p178618	Giorgia Polimeni	NaN	NaN	NaN	Italy	
2	p178616	Silvia Simone	NaN	NaN	NaN	Italy	
3	p178614	Martina Mazza	NaN	NaN	NaN	Italy	
4	p178613	Sophia Letizia Foti	NaN	NaN	NaN	Italy	

	Position	Dominant hand	Height (cm)	Weight (kg)	...	Followers	\
0	Universal	NaN	NaN	NaN	...	NaN	
1	Universal	NaN	NaN	NaN	...	NaN	
2	Universal	NaN	NaN	NaN	...	NaN	
3	Universal	NaN	NaN	NaN	...	NaN	
4	Universal	NaN	NaN	NaN	...	NaN	

	Player ranking	Coach ranking	Registered	Views	Last activity	\
0	NaN	NaN	No	6	NaN	
1	NaN	NaN	No	6	NaN	
2	NaN	NaN	No	6	NaN	
3	NaN	NaN	No	6	NaN	
4	NaN	NaN	No	6	NaN	

Added by User Recommendations Partners

0	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN
1	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN
2	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN
3	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN
4	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN

[5 rows x 50 columns]

```
[16]: dedup_len = len(player_dedup)
dedup_len
```

[16]: 78549

1.1.2 Second Transformation

```
[19]: # Reduce dataset to remove retired players based on age > 50
# Left NULL values in as some are current players
player_age = player_dedup[(player_dedup['Birthdate'] >= '1975-01-01') |
↪ (player_dedup['Birthdate'].isnull())]
print(player_age.head())
```

	Id	Name	Nickname	Maiden name	Birthdate	Nationality	\
0	p178623	Giada Maria Falsone	NaN	NaN	NaN	Italy	
1	p178618	Giorgia Polimeni	NaN	NaN	NaN	Italy	
2	p178616	Silvia Simone	NaN	NaN	NaN	Italy	
3	p178614	Martina Mazza	NaN	NaN	NaN	Italy	
4	p178613	Sophia Letizia Foti	NaN	NaN	NaN	Italy	

	Position	Dominant hand	Height (cm)	Weight (kg)	...	Followers	\
0	Universal	NaN	NaN	NaN	...	NaN	
1	Universal	NaN	NaN	NaN	...	NaN	
2	Universal	NaN	NaN	NaN	...	NaN	
3	Universal	NaN	NaN	NaN	...	NaN	
4	Universal	NaN	NaN	NaN	...	NaN	

	Player ranking	Coach ranking	Registered	Views	Last activity	\
0	NaN	NaN	No	6	NaN	
1	NaN	NaN	No	6	NaN	
2	NaN	NaN	No	6	NaN	
3	NaN	NaN	No	6	NaN	
4	NaN	NaN	No	6	NaN	

	Added by	User	Recommendations	Partners
0	Mattia	Mario	Guido	Caputo
1	Mattia	Mario	Guido	Caputo
2	Mattia	Mario	Guido	Caputo
3	Mattia	Mario	Guido	Caputo
4	Mattia	Mario	Guido	Caputo

[5 rows x 50 columns]

```
[21]: age_len = len(player_age)
      age_len
```

[21]: 76573

1.1.3 Third Transformation

```
[24]: # Reorder of columns in data set to put key ones first
player_reorder = player_age[['Id', 'Name', 'Birthdate', 'Nationality',
    ↪ 'Position', 'Dominant hand', 'Height (cm)', 'Weight (kg)', 'Season',
    'Club', 'ClubId', 'LeagueCountry', 'ActivePlayer', 'Clubs', 'Teammates',
    ↪ 'Tournaments', 'Matches', 'Awards',
    'Ranking', 'Ranking in 2023', 'Player ranking', 'Coach ranking', 'Registered',
    ↪ 'Views', 'Nickname', 'Maiden name',
    'Spike (cm)', 'Block (cm)', 'Url', 'Image', 'ClubUrl', 'ClubLogo',
    ↪ 'LeagueCountryFlag', 'Agent', 'Top100', 'Top200',
    'Top500', 'Twitter', 'Facebook', 'Instagram', 'Youtube', 'Linkedin', 'Website',
    ↪ 'Timeline', 'Followers', 'Last activity',
    'Added by', 'User', 'Recommendations', 'Partners']]
player_reorder.head()
```

```
[24]:
```

		Id	Name	Birthdate	Nationality	Position	\
0	p178623	Giada Maria Falsone	NaN	Italy	Universal		
1	p178618	Giorgia Polimeni	NaN	Italy	Universal		
2	p178616	Silvia Simone	NaN	Italy	Universal		
3	p178614	Martina Mazza	NaN	Italy	Universal		
4	p178613	Sophia Letizia Foti	NaN	Italy	Universal		

		Dominant hand	Height (cm)	Weight (kg)	Season	\
0		NaN	NaN	NaN	2023/24 - 2023/24	
1		NaN	NaN	NaN	2023/24 - 2023/24	
2		NaN	NaN	NaN	2023/24 - 2023/24	
3		NaN	NaN	NaN	2023/24 - 2023/24	
4		NaN	NaN	NaN	2023/24 - 2023/24	

		Club	...	Youtube	Linkedin	Website	\
0	Polisportiva Elio Sozzi Reggio Calabria	...		NaN	NaN	NaN	
1	Polisportiva Elio Sozzi Reggio Calabria	...		NaN	NaN	NaN	
2	Polisportiva Elio Sozzi Reggio Calabria	...		NaN	NaN	NaN	
3	Polisportiva Elio Sozzi Reggio Calabria	...		NaN	NaN	NaN	
4	Polisportiva Elio Sozzi Reggio Calabria	...		NaN	NaN	NaN	

		Timeline	Followers	Last activity	Added by	User	\
0	0	NaN	NaN	Mattia Mario Guido Caputo	NaN		

1	0	NaN	NaN	Mattia	Mario	Guido	Caputo	NaN
2	0	NaN	NaN	Mattia	Mario	Guido	Caputo	NaN
3	0	NaN	NaN	Mattia	Mario	Guido	Caputo	NaN
4	0	NaN	NaN	Mattia	Mario	Guido	Caputo	NaN

	Recommendations	Partners
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN

[5 rows x 50 columns]

1.1.4 Fourth Transformation

```
[27]: # Correct Players Height/Weight
```

```
[29]: # Alice Gillier's height should be 180 not 280
player_reorder.loc[player_reorder['Name'] == 'Alice Gillier', 'Height (cm)'] = 180
```

```
[31]: # Null out weight above 140 kg
player_reorder.loc[player_reorder['Weight (kg)'] >= 140] = np.nan
player_reorder['Weight (kg)'].unique()
```

```
[31]: array([ nan,  68.,  55.,  65.,  58.,  50.,  60.,  62.,  66.,  61.,  70.,
         67.,  80.,  75.,  54.,  63.,  49.,  56.,  59.,  64.,  57.,  72.,
         69.,  53.,  51.,  96.,  78.,  85.,  79.,  74.,  52.,  89.,  71.,
         46.,  86.,  87.,  48.,  97.,  95.,  76.,  73.,  77.,  83.,  47.,
         81.,  84.,  82.,  44.,  45.,  90.,  38.,  43.,  92.,  91.,  88.,
         98., 110., 104.,  93., 105.,  99., 102.,  94., 100., 111.,  40.,
         42.,  41., 120.,  39.,  35.] )
```

1.1.5 Fifth Transformation

```
[34]: # Convert Height from centimeters to inches
player_reorder['Height (cm)'] = player_reorder['Height (cm)'] * 0.393701
player_reorder.loc[:, 'Height (cm)']
```

```
[34]: 0      NaN
      1      NaN
      2      NaN
      3      NaN
      4      NaN
      ...
      78687  68.897675
      78689  66.141768
```

```

78690    71.259881
78691         NaN
78692         NaN
Name: Height (cm), Length: 76573, dtype: float64

```

```

[36]: # Rename field to indicate correct measurement type
player_reorder = player_reorder.rename(columns={'Height (cm)': 'Height (in)'})
player_reorder.head()

```

```

[36]:      Id      Name Birthdate Nationality  Position \
0  p178623  Giada Maria Falsone      NaN      Italy  Universal
1  p178618   Giorgia Polimeni      NaN      Italy  Universal
2  p178616   Silvia Simone      NaN      Italy  Universal
3  p178614   Martina Mazza      NaN      Italy  Universal
4  p178613  Sophia Letizia Foti      NaN      Italy  Universal

      Dominant hand  Height (in)  Weight (kg)      Season \
0             NaN          NaN          NaN  2023/24 - 2023/24
1             NaN          NaN          NaN  2023/24 - 2023/24
2             NaN          NaN          NaN  2023/24 - 2023/24
3             NaN          NaN          NaN  2023/24 - 2023/24
4             NaN          NaN          NaN  2023/24 - 2023/24

      Club  ... Youtube LinkedIn Website \
0  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN
1  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN
2  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN
3  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN
4  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN

      Timeline  Followers  Last activity      Added by  User \
0          0.0          NaN          NaN  Mattia Mario Guido Caputo  NaN
1          0.0          NaN          NaN  Mattia Mario Guido Caputo  NaN
2          0.0          NaN          NaN  Mattia Mario Guido Caputo  NaN
3          0.0          NaN          NaN  Mattia Mario Guido Caputo  NaN
4          0.0          NaN          NaN  Mattia Mario Guido Caputo  NaN

      Recommendations  Partners
0             NaN          NaN
1             NaN          NaN
2             NaN          NaN
3             NaN          NaN
4             NaN          NaN

```

```

[5 rows x 50 columns]

```

1.1.6 Sixth Transformation

```
[38]: # Convert Weight from kilograms to pounds
player_reorder['Weight (kg)'] = player_reorder['Weight (kg)'] * 2.20462
player_reorder.loc[:, 'Weight (kg)']
```

```
[38]: 0          NaN
      1          NaN
      2          NaN
      3          NaN
      4          NaN
      ...
      78687    156.52802
      78689    130.07258
      78690    147.70954
      78691          NaN
      78692          NaN
      Name: Weight (kg), Length: 76573, dtype: float64
```

```
[40]: # Rename field to indicate correct measurement type
player_reorder = player_reorder.rename(columns={'Weight (kg)': 'Weight (lb)'})
player_reorder.head()
```

```
[40]:      Id      Name Birthdate Nationality  Position \
0  p178623  Giada Maria Falsone      NaN      Italy  Universal
1  p178618   Giorgia Polimeni      NaN      Italy  Universal
2  p178616   Silvia Simone      NaN      Italy  Universal
3  p178614   Martina Mazza      NaN      Italy  Universal
4  p178613  Sophia Letizia Foti      NaN      Italy  Universal

      Dominant hand  Height (in)  Weight (lb)      Season \
0          NaN          NaN          NaN  2023/24 - 2023/24
1          NaN          NaN          NaN  2023/24 - 2023/24
2          NaN          NaN          NaN  2023/24 - 2023/24
3          NaN          NaN          NaN  2023/24 - 2023/24
4          NaN          NaN          NaN  2023/24 - 2023/24

      Club  ... Youtube LinkedIn Website \
0  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN
1  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN
2  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN
3  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN
4  Polisportiva Elio Sozzi Reggio Calabria  ...      NaN      NaN      NaN

      Timeline  Followers  Last activity      Added by  User \
0          0.0          NaN          NaN  Mattia Mario Guido Caputo  NaN
1          0.0          NaN          NaN  Mattia Mario Guido Caputo  NaN
2          0.0          NaN          NaN  Mattia Mario Guido Caputo  NaN
```

3	0.0	NaN	NaN	Mattia	Mario	Guido	Caputo	NaN
4	0.0	NaN	NaN	Mattia	Mario	Guido	Caputo	NaN

	Recommendations	Partners
0	NaN	NaN
1	NaN	NaN
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN

[5 rows x 50 columns]

1.1.7 Seventh Transformation

```
[43]: # Remove "-" from Position Field
player_reorder['Position'] = player_reorder['Position'].replace({'-': ' ', })
player_reorder['Position'].unique()
```

```
[43]: array(['Universal', 'Middle-blocker', 'Setter', 'Libero',
        'Outside Hitter', 'Opposite', nan], dtype=object)
```

```
[45]: # Show Final Table
pd.set_option('display.max_columns', None)
player_reorder.head()
```

```
[45]:      Id      Name Birthdate Nationality  Position \
0  p178623  Giada Maria Falsone      NaN      Italy  Universal
1  p178618   Giorgia Polimeni      NaN      Italy  Universal
2  p178616   Silvia Simone      NaN      Italy  Universal
3  p178614   Martina Mazza      NaN      Italy  Universal
4  p178613  Sophia Letizia Foti      NaN      Italy  Universal
```

	Dominant hand	Height (in)	Weight (lb)	Season \
0	NaN	NaN	NaN	2023/24 - 2023/24
1	NaN	NaN	NaN	2023/24 - 2023/24
2	NaN	NaN	NaN	2023/24 - 2023/24
3	NaN	NaN	NaN	2023/24 - 2023/24
4	NaN	NaN	NaN	2023/24 - 2023/24

	Club	ClubId	LeagueCountry	ActivePlayer \
0	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	Yes
1	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	Yes
2	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	Yes
3	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	Yes
4	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	Yes

Clubs	Teammates	Tournaments	Matches	Awards	Ranking	Ranking in 2023 \
-------	-----------	-------------	---------	--------	---------	-------------------

0	1.0	16.0	1.0	0.0	0.0	NaN	NaN
1	1.0	16.0	1.0	0.0	0.0	NaN	NaN
2	1.0	16.0	1.0	0.0	0.0	NaN	NaN
3	1.0	16.0	1.0	0.0	0.0	NaN	NaN
4	1.0	16.0	1.0	0.0	0.0	NaN	NaN

	Player ranking	Coach ranking	Registered	Views	Nickname	Maiden name	\
0	NaN	NaN	No	6.0	NaN	NaN	
1	NaN	NaN	No	6.0	NaN	NaN	
2	NaN	NaN	No	6.0	NaN	NaN	
3	NaN	NaN	No	6.0	NaN	NaN	
4	NaN	NaN	No	6.0	NaN	NaN	

	Spike (cm)	Block (cm)	Url	\
0	NaN	NaN	https://women.volleybox.net/giada-maria-falson...	
1	NaN	NaN	https://women.volleybox.net/giorgia-polimeni-p...	
2	NaN	NaN	https://women.volleybox.net/silvia-simone-p178616	
3	NaN	NaN	https://women.volleybox.net/martina-mazza-p178614	
4	NaN	NaN	https://women.volleybox.net/sophia-letizia-fot...	

	Image	\
0	https://volleybox.net/media/upload/players/def...	
1	https://volleybox.net/media/upload/players/def...	
2	https://volleybox.net/media/upload/players/def...	
3	https://volleybox.net/media/upload/players/def...	
4	https://volleybox.net/media/upload/players/def...	

	ClubUrl	\
0	https://women.volleybox.net/polisportiva-elio-...	
1	https://women.volleybox.net/polisportiva-elio-...	
2	https://women.volleybox.net/polisportiva-elio-...	
3	https://women.volleybox.net/polisportiva-elio-...	
4	https://women.volleybox.net/polisportiva-elio-...	

	ClubLogo	\
0	https://volleybox.net/media/upload/teams/16938...	
1	https://volleybox.net/media/upload/teams/16938...	
2	https://volleybox.net/media/upload/teams/16938...	
3	https://volleybox.net/media/upload/teams/16938...	
4	https://volleybox.net/media/upload/teams/16938...	

	LeagueCountryFlag	Agent	Top100	Top200	Top500	\
0	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No	
1	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No	
2	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No	
3	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No	
4	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No	

	Twitter	Facebook	Instagram	Youtube	Linkedin	Website	Timeline	Followers	\
0	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	
1	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	
2	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	
3	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	
4	NaN	NaN	NaN	NaN	NaN	NaN	0.0	NaN	

	Last activity	Added by	User	Recommendations	Partners
0	NaN	Mattia Mario Guido Caputo	NaN	NaN	NaN
1	NaN	Mattia Mario Guido Caputo	NaN	NaN	NaN
2	NaN	Mattia Mario Guido Caputo	NaN	NaN	NaN
3	NaN	Mattia Mario Guido Caputo	NaN	NaN	NaN
4	NaN	Mattia Mario Guido Caputo	NaN	NaN	NaN

1.1.8 Eighth Transformation

```
[48]: # Splitting player name into first and last name
player_reorder['FName'] = player_reorder['Name'].str.split().str[0]
player_reorder['Lname'] = player_reorder['Name'].str.split().str[-1]
player_reorder['first_name'] = player_reorder['Name'].str.split().str[0]
player_reorder['last_name'] = player_reorder['Name'].str.split().str[1:].str.
    ↪join(' ')
player_reorder['middle_name'] = player_reorder['last_name'].str.split().str[0]
player_reorder['last_name_final'] = player_reorder['last_name'].str.split().
    ↪str[1:].str.join(' ')
player_reorder['last_name_final'] = player_reorder.apply(lambda x:
    ↪x['middle_name'] if x['last_name_final']==' else x['last_name_final'],
    ↪axis=1)
player_reorder['middle_name'] = player_reorder.apply(lambda x: x['Lname'] if
    ↪x['last_name_final']==x['middle_name'] else x['middle_name'], axis=1)
player_reorder['first_initial'] = player_reorder.first_name.str[0]
```

```
[50]: # Drop extra name fields
del player_reorder['FName']
del player_reorder['Lname']
```

```
[52]: print(player_reorder.head(40))
```

	Id	Name	Birthdate	Nationality	Position	\
0	p178623	Giada Maria Falsone	NaN	Italy	Universal	
1	p178618	Giorgia Polimeni	NaN	Italy	Universal	
2	p178616	Silvia Simone	NaN	Italy	Universal	
3	p178614	Martina Mazza	NaN	Italy	Universal	
4	p178613	Sophia Letizia Foti	NaN	Italy	Universal	
5	p178612	Martina Caruso	NaN	Italy	Universal	
6	p178611	Reina Hamano	1976-07-19	Japan	Middle-blocker	

7	p178610	Erika Furfaro	NaN	Italy	Universal
8	p178609	Lara Battimelli	NaN	Italy	Universal
9	p178604	Irene Pizzarello	2000-01-01	Italy	Setter
10	p178603	Giada Calluso	2008-01-01	Italy	Universal
11	p178602	Giorgia Occhiuto	2008-01-01	Italy	Universal
12	p178598	Rosana Pryston	2009-08-28	Brazil	Middle-blocker
13	p178597	Nguyễn Thị Phước	1996-01-01	Vietnam	Libero
14	p178595	Zeynep Kılıç	2012-04-17	Turkey	Outside Hitter
15	p178594	Ioana Cuciurean	NaN	Romania	Outside Hitter
16	p178592	Ilinca Boboc	NaN	Romania	Setter
17	p178591	Francesca Hînganu Marian	NaN	Romania	Outside Hitter
18	p178588	Maria Chiara Manca	NaN	Italy	Opposite
19	p178587	Maria Luisa Pezzuto	NaN	Italy	Middle-blocker
20	p178586	Sara Quarta	NaN	Italy	Setter
21	p178585	Federica Albano	2005-01-01	Italy	Outside Hitter
22	p178584	Benedetta Capoccia	NaN	Italy	Outside Hitter
23	p178582	Sara Toto	NaN	Italy	Universal
24	p178579	Giulia Agostini	2002-11-11	Italy	Setter
25	p178573	Mariasara Guido	NaN	Italy	Setter
26	p178572	Vittoria Ciavatta	NaN	Italy	Universal
27	p178571	Giulia Marcattili	NaN	Italy	Libero
28	p178570	Corinne Perna	NaN	Italy	Libero
29	p178569	Emma Oldani	NaN	Italy	Universal
30	p178568	Lara Benedetta Ianni	2009-01-01	Italy	Opposite
31	p178566	Francesca Mattarollo	2002-01-01	Italy	Middle-blocker
32	p178565	Annapaola Lombardo	1981-01-01	Italy	Middle-blocker
33	p178564	Claudia Rotili	2007-01-01	Italy	Middle-blocker
34	p178563	Chiara Cavalli	1993-01-01	Italy	Middle-blocker
35	p178562	Sara Gianfrancesco	1998-01-01	Italy	Libero
36	p178561	Sara Elleboro	1991-01-01	Italy	Libero
37	p178557	Alessia Carola Salvia	NaN	Italy	Universal
38	p178555	Federica Calzonetti	NaN	Italy	Middle-blocker
39	p178554	Alessia De Marco	NaN	Italy	Universal

	Dominant hand	Height (in)	Weight (lb)	Season \
0	NaN	NaN	NaN	2023/24 - 2023/24
1	NaN	NaN	NaN	2023/24 - 2023/24
2	NaN	NaN	NaN	2023/24 - 2023/24
3	NaN	NaN	NaN	2023/24 - 2023/24
4	NaN	NaN	NaN	2023/24 - 2023/24
5	NaN	NaN	NaN	2023/24 - 2023/24
6	Right	70.866180	149.91416	1999/00 - 2003/04
7	NaN	NaN	NaN	2023/24 - 2023/24
8	NaN	NaN	NaN	2023/24 - 2023/24
9	NaN	NaN	NaN	2023/24 - 2023/24
10	NaN	NaN	NaN	2023/24 - 2023/24
11	NaN	NaN	NaN	2023/24 - 2023/24
12	Right	74.803190	NaN	2023/24 - 2023/24

13	Right	65.748067	121.25410	2022/23 - 2022/23
14	Right	70.078778	143.30030	2023/24 - still
15	NaN	NaN	NaN	2023/24 - 2023/24
16	NaN	NaN	NaN	2023/24 - 2023/24
17	NaN	NaN	NaN	2023/24 - 2023/24
18	NaN	NaN	NaN	2023/24 - 2023/24
19	NaN	NaN	NaN	2023/24 - 2023/24
20	NaN	NaN	NaN	2023/24 - 2023/24
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39	NaN	NaN	NaN	2023/24 - 2023/24

	Club	ClubId	LeagueCountry	\
0	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	
1	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	
2	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	
3	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	
4	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	
5	Polisportiva Elio Sozzi Reggio Calabria	t20329	Italy	
6	Denso Airybees	t2090	Japan	
7	Pink Volo Lamezia	t24983	Italy	
8	Pink Volo Lamezia	t24983	Italy	
9	New Teosidos Reggio Calabria	t19667	Italy	
10	New Teosidos Reggio Calabria	t19667	Italy	
11	New Teosidos Reggio Calabria	t19667	Italy	
12	Recife Vôlei	t2934	Brazil	
13	Hà Phú Thanh Hóa	t6941	Vietnam	
14	Vakıfbank II	t7387	Turkey	
15	CS Știința Bacău	t2280	Romania	
16	CS Știința Bacău	t2280	Romania	
17	CS Știința Bacău	t2280	Romania	
18	Trepuzzi Volley	t23429	Italy	

19		Trepuzzi Volley	t23429	Italy
20		Trepuzzi Volley	t23429	Italy
21		Trepuzzi Volley	t23429	Italy
22		Trepuzzi Volley	t23429	Italy
23	CiDue Costruzioni Snc New Fides Volley Campo C...		t24309	Italy
24		Solteri Volley Trento	t19815	Italy
25		Leo Constructions Monteroni	t23535	Italy
26		Viva Volley Tortoreto	t6080	Italy
27		Viva Volley Tortoreto	t6080	Italy
28		Viva Volley Tortoreto	t6080	Italy
29		Studio Industria Volley L'Aquila	t24942	Italy
30		Studio Industria Volley L'Aquila	t24942	Italy
31		Studio Industria Volley L'Aquila	t24942	Italy
32		Studio Industria Volley L'Aquila	t24942	Italy
33		Studio Industria Volley L'Aquila	t24942	Italy
34		Studio Industria Volley L'Aquila	t24942	Italy
35		Studio Industria Volley L'Aquila	t24942	Italy
36		Studio Industria Volley L'Aquila	t24942	Italy
37		Polisportiva Francavilla	t24939	Italy
38		Polisportiva Francavilla	t24939	Italy
39		Polisportiva Francavilla	t24939	Italy

	ActivePlayer	Clubs	Teammates	Tournaments	Matches	Awards	Ranking \
0	Yes	1.0	16.0	1.0	0.0	0.0	NaN
1	Yes	1.0	16.0	1.0	0.0	0.0	NaN
2	Yes	1.0	16.0	1.0	0.0	0.0	NaN
3	Yes	1.0	16.0	1.0	0.0	0.0	NaN
4	Yes	1.0	16.0	1.0	0.0	0.0	NaN
5	Yes	1.0	16.0	1.0	0.0	0.0	NaN
6	No	4.0	24.0	21.0	12.0	0.0	NaN
7	Yes	1.0	13.0	1.0	0.0	0.0	NaN
8	Yes	1.0	13.0	1.0	0.0	0.0	NaN
9	Yes	1.0	17.0	1.0	0.0	0.0	NaN
10	Yes	1.0	17.0	1.0	0.0	0.0	NaN
11	Yes	1.0	17.0	1.0	0.0	0.0	NaN
12	Yes	1.0	1.0	1.0	0.0	0.0	NaN
13	No	1.0	12.0	2.0	11.0	0.0	NaN
14	Yes	4.0	32.0	1.0	4.0	0.0	NaN
15	Yes	1.0	13.0	1.0	4.0	0.0	NaN
16	Yes	1.0	13.0	1.0	4.0	0.0	NaN
17	Yes	1.0	13.0	1.0	4.0	0.0	NaN
18	Yes	1.0	14.0	1.0	0.0	0.0	NaN
19	Yes	1.0	14.0	1.0	0.0	0.0	NaN
20	Yes	1.0	14.0	1.0	0.0	0.0	NaN
21	Yes	1.0	14.0	1.0	0.0	0.0	NaN
22	Yes	1.0	14.0	1.0	0.0	0.0	NaN
23	Yes	1.0	15.0	1.0	0.0	0.0	NaN
24	Yes	1.0	4.0	1.0	0.0	0.0	NaN

25	Yes	1.0	12.0	1.0	0.0	0.0	NaN
26	Yes	1.0	16.0	1.0	0.0	0.0	NaN
27	Yes	1.0	16.0	1.0	0.0	0.0	NaN
28	Yes	1.0	16.0	1.0	0.0	0.0	NaN
29	Yes	1.0	15.0	1.0	0.0	0.0	NaN
30	Yes	1.0	15.0	1.0	0.0	0.0	NaN
31	Yes	1.0	15.0	1.0	0.0	0.0	NaN
32	Yes	1.0	15.0	1.0	0.0	0.0	NaN
33	Yes	1.0	15.0	1.0	0.0	0.0	NaN
34	Yes	1.0	15.0	1.0	0.0	0.0	NaN
35	Yes	1.0	15.0	1.0	0.0	0.0	NaN
36	Yes	1.0	15.0	1.0	0.0	0.0	NaN
37	Yes	1.0	15.0	1.0	0.0	0.0	NaN
38	Yes	1.0	15.0	1.0	0.0	0.0	NaN
39	Yes	1.0	15.0	1.0	0.0	0.0	NaN

	Ranking in 2023	Player ranking	Coach ranking	Registered	Views \
0	NaN	NaN	NaN	No	6.0
1	NaN	NaN	NaN	No	6.0
2	NaN	NaN	NaN	No	6.0
3	NaN	NaN	NaN	No	6.0
4	NaN	NaN	NaN	No	6.0
5	NaN	NaN	NaN	No	6.0
6	NaN	NaN	NaN	No	9.0
7	NaN	NaN	NaN	No	18.0
8	NaN	NaN	NaN	No	13.0
9	NaN	NaN	NaN	No	16.0
10	NaN	NaN	NaN	No	14.0
11	NaN	NaN	NaN	No	13.0
12	NaN	NaN	NaN	No	29.0
13	NaN	NaN	NaN	No	13.0
14	NaN	NaN	NaN	No	33.0
15	NaN	NaN	NaN	No	15.0
16	NaN	NaN	NaN	No	17.0
17	NaN	NaN	NaN	No	6.0
18	NaN	NaN	NaN	No	13.0
19	NaN	NaN	NaN	No	7.0
20	NaN	NaN	NaN	No	8.0
21	NaN	NaN	NaN	No	6.0
22	NaN	NaN	NaN	No	8.0
23	NaN	NaN	NaN	No	5.0
24	NaN	NaN	NaN	No	11.0
25	NaN	NaN	NaN	No	12.0
26	NaN	NaN	NaN	No	5.0
27	NaN	NaN	NaN	No	5.0
28	NaN	NaN	NaN	No	8.0
29	NaN	NaN	NaN	No	6.0
30	NaN	NaN	NaN	No	7.0

31	NaN	NaN	NaN	No	7.0
32	NaN	NaN	NaN	No	9.0
33	NaN	NaN	NaN	No	9.0
34	NaN	NaN	NaN	No	7.0
35	NaN	NaN	NaN	No	6.0
36	NaN	NaN	NaN	No	5.0
37	NaN	NaN	NaN	No	4.0
38	NaN	NaN	NaN	No	5.0
39	NaN	NaN	NaN	No	4.0

	Nickname	Maiden name	Spike (cm)	Block (cm)	\
0	NaN	NaN	NaN	NaN	
1	NaN	NaN	NaN	NaN	
2	NaN	NaN	NaN	NaN	
3	NaN	NaN	NaN	NaN	
4	NaN	NaN	NaN	NaN	
5	NaN	NaN	NaN	NaN	
6	NaN		305.0	300.0	
7	NaN	NaN	NaN	NaN	
8	NaN	NaN	NaN	NaN	
9	NaN	NaN	NaN	NaN	
10	NaN	NaN	NaN	NaN	
11	NaN	NaN	NaN	NaN	
12	NaN	NaN	NaN	NaN	
13	NaN	NaN	250.0	235.0	
14	Malatyali	NaN	281.0	NaN	
15	NaN	NaN	NaN	NaN	
16	NaN	NaN	NaN	NaN	
17	NaN	NaN	NaN	NaN	
18	NaN	NaN	NaN	NaN	
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27	NaN	NaN	NaN	NaN	
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29	NaN	NaN	NaN	NaN	
30	NaN	NaN	NaN	NaN	
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35	NaN	NaN	NaN	NaN	
36	NaN	NaN	NaN	NaN	

37	NaN	NaN	NaN	NaN
38	NaN	NaN	NaN	NaN
39	NaN	NaN	NaN	NaN

Url \

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2 https://women.volleybox.net/silvia-simone-p178616
3 https://women.volleybox.net/martina-mazza-p178614
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5 https://women.volleybox.net/martina-caruso-p17...
6 https://women.volleybox.net/reina-hamano-p178611
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9 https://women.volleybox.net/irene-pizzarello-p...
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16 https://women.volleybox.net/ilinca-boboc-p178592
17 https://women.volleybox.net/francesca-hinganu-...
18 https://women.volleybox.net/maria-chiara-manca...
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21 https://women.volleybox.net/federica-albano-p1...
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Image \

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ClubLogo \

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3	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No	
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5	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No	
6	https://volleybox.net/media/img/flags/JP.png	NaN	No	No	No	
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26	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
27	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
28	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
29	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
30	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
31	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
32	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
33	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
34	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
35	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
36	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
37	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
38	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No
39	https://volleybox.net/media/img/flags/IT.png	NaN	No	No	No

	Twitter	Facebook		Instagram	Youtube	\
0	NaN	NaN		NaN	NaN	
1	NaN	NaN		NaN	NaN	
2	NaN	NaN		NaN	NaN	
3	NaN	NaN		NaN	NaN	
4	NaN	NaN		NaN	NaN	
5	NaN	NaN		NaN	NaN	
6	NaN	NaN		NaN	NaN	
7	NaN	NaN		NaN	NaN	
8	NaN	NaN		NaN	NaN	
9	NaN	NaN		NaN	NaN	
10	NaN	NaN		NaN	NaN	
11	NaN	NaN		NaN	NaN	
12	NaN	NaN	https://www.instagram.com/rosanaprysthon/		NaN	
13	NaN	NaN		NaN	NaN	
14	NaN	NaN		NaN	NaN	
15	NaN	NaN		NaN	NaN	
16	NaN	NaN		NaN	NaN	
17	NaN	NaN		NaN	NaN	
18	NaN	NaN		NaN	NaN	
19	NaN	NaN		NaN	NaN	
20	NaN	NaN		NaN	NaN	
21	NaN	NaN		NaN	NaN	
22	NaN	NaN		NaN	NaN	
23	NaN	NaN		NaN	NaN	
24	NaN	NaN		NaN	NaN	

25	NaN	NaN			NaN	NaN
26	NaN	NaN			NaN	NaN
27	NaN	NaN			NaN	NaN
28	NaN	NaN			NaN	NaN
29	NaN	NaN			NaN	NaN
30	NaN	NaN			NaN	NaN
31	NaN	NaN			NaN	NaN
32	NaN	NaN			NaN	NaN
33	NaN	NaN			NaN	NaN
34	NaN	NaN			NaN	NaN
35	NaN	NaN			NaN	NaN
36	NaN	NaN			NaN	NaN
37	NaN	NaN			NaN	NaN
38	NaN	NaN			NaN	NaN
39	NaN	NaN			NaN	NaN

	Linkedin	Website	Timeline	Followers	Last activity	\
0	NaN	NaN	0.0	NaN	NaN	
1	NaN	NaN	0.0	NaN	NaN	
2	NaN	NaN	0.0	NaN	NaN	
3	NaN	NaN	0.0	NaN	NaN	
4	NaN	NaN	0.0	NaN	NaN	
5	NaN	NaN	0.0	NaN	NaN	
6	NaN	NaN	0.0	NaN	NaN	
7	NaN	NaN	0.0	NaN	NaN	
8	NaN	NaN	0.0	NaN	NaN	
9	NaN	NaN	0.0	NaN	NaN	
10	NaN	NaN	0.0	NaN	NaN	
11	NaN	NaN	0.0	NaN	NaN	
12	NaN	NaN	0.0	NaN	NaN	
13	NaN	NaN	0.0	NaN	NaN	
14	NaN	NaN	1.0	NaN	2 minutes ago	
15	NaN	NaN	0.0	NaN	NaN	
16	NaN	NaN	0.0	NaN	NaN	
17	NaN	NaN	0.0	NaN	NaN	
18	NaN	NaN	0.0	NaN	NaN	
19	NaN	NaN	0.0	NaN	NaN	
20	NaN	NaN	0.0	NaN	NaN	
21	NaN	NaN	0.0	NaN	NaN	
22	NaN	NaN	0.0	NaN	NaN	
23	NaN	NaN	0.0	NaN	NaN	
24	NaN	NaN	0.0	NaN	NaN	
25	NaN	NaN	0.0	NaN	NaN	
26	NaN	NaN	0.0	NaN	NaN	
27	NaN	NaN	0.0	NaN	NaN	
28	NaN	NaN	0.0	NaN	NaN	
29	NaN	NaN	0.0	NaN	NaN	
30	NaN	NaN	0.0	NaN	NaN	

31	NaN	NaN	0.0	NaN	NaN
32	NaN	NaN	0.0	NaN	NaN
33	NaN	NaN	0.0	NaN	NaN
34	NaN	NaN	0.0	NaN	NaN
35	NaN	NaN	0.0	NaN	NaN
36	NaN	NaN	0.0	NaN	NaN
37	NaN	NaN	0.0	NaN	NaN
38	NaN	NaN	0.0	NaN	NaN
39	NaN	NaN	0.0	NaN	NaN

	Added by				User	Recommendations	Partners	first_name	\
0	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Giada	
1	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Giorgia	
2	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Silvia	
3	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Martina	
4	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Sophia	
5	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Martina	
6			sitenoise		NaN	NaN	NaN	Reina	
7	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Erika	
8	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Lara	
9	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Irene	
10	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Giada	
11	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Giorgia	
12			Abramovitz		NaN	NaN	NaN	Rosana	
13			Nguyen Tai		NaN	NaN	NaN	Nguyễn	
14			NaN		NaN	NaN	NaN	Zeynep	
15			Sorin Acela		NaN	NaN	NaN	Ioana	
16			Sorin Acela		NaN	NaN	NaN	Ilinca	
17			Sorin Acela		NaN	NaN	NaN	Francesca	
18			fast_44		NaN	NaN	NaN	Maria	
19			fast_44		NaN	NaN	NaN	Maria	
20			fast_44		NaN	NaN	NaN	Sara	
21			fast_44		NaN	NaN	NaN	Federica	
22			fast_44		NaN	NaN	NaN	Benedetta	
23	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Sara	
24	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Giulia	
25			fast_44		NaN	NaN	NaN	Mariasara	
26	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Vittoria	
27	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Giulia	
28	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Corinne	
29	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Emma	
30	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Lara	
31	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Francesca	
32	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Annapaola	
33	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Claudia	
34	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Chiara	
35	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Sara	
36	Mattia	Mario	Guido	Caputo	NaN	NaN	NaN	Sara	

37	Mattia Mario Guido Caputo	NaN	NaN	NaN	Alessia
38	Mattia Mario Guido Caputo	NaN	NaN	NaN	Federica
39	Mattia Mario Guido Caputo	NaN	NaN	NaN	Alessia

	last_name	middle_name	last_name_final	first_initial
0	Maria Falsone	Maria	Falsone	G
1	Polimeni	Polimeni	Polimeni	G
2	Simone	Simone	Simone	S
3	Mazza	Mazza	Mazza	M
4	Letizia Foti	Letizia	Foti	S
5	Caruso	Caruso	Caruso	M
6	Hamano	Hamano	Hamano	R
7	Furfaro	Furfaro	Furfaro	E
8	Battimelli	Battimelli	Battimelli	L
9	Pizzarello	Pizzarello	Pizzarello	I
10	Calluso	Calluso	Calluso	G
11	Occhiuto	Occhiuto	Occhiuto	G
12	Pryston	Pryston	Pryston	R
13	Thị Phước	Thị	Phước	N
14	Kılıç	Kılıç	Kılıç	Z
15	Cuciurean	Cuciurean	Cuciurean	I
16	Boboc	Boboc	Boboc	I
17	Hînganu Marian	Hînganu	Marian	F
18	Chiara Manca	Chiara	Manca	M
19	Luisa Pezzuto	Luisa	Pezzuto	M
20	Quarta	Quarta	Quarta	S
21	Albano	Albano	Albano	F
22	Capoccia	Capoccia	Capoccia	B
23	Toto	Toto	Toto	S
24	Agostini	Agostini	Agostini	G
25	Guido	Guido	Guido	M
26	Ciavatta	Ciavatta	Ciavatta	V
27	Marcattili	Marcattili	Marcattili	G
28	Perna	Perna	Perna	C
29	Oldani	Oldani	Oldani	E
30	Benedetta Ianni	Benedetta	Ianni	L
31	Mattarollo	Mattarollo	Mattarollo	F
32	Lombardo	Lombardo	Lombardo	A
33	Rotili	Rotili	Rotili	C
34	Cavalli	Cavalli	Cavalli	C
35	Gianfrancesco	Gianfrancesco	Gianfrancesco	S
36	Elleboro	Elleboro	Elleboro	S
37	Carola Salvia	Carola	Salvia	A
38	Calzonetti	Calzonetti	Calzonetti	F
39	De Marco	De	Marco	A

There were very few changes to the data itself. The height and weight fields were converted into standard from metric.

The one height outlier was looked up and changed. The weight outliers were changed to NaN. These conversions provide very little risk as the outliers were removed and typically weight does not have a major impact to player performance. Since Kaggle was my source for data and users can upload or correct the data on Kaggle, there is the chance that data could be entered incorrectly. I sampled a well known professional volleyball player from Nebraska (Jordan Larson) and the data for her was accurate. Finding a way to get player statistics direct from the professional leagues would be a way to make the data more credible, but may take more work if there isn't a unified dataset provided by the leagues because you'd have to find stats on each team.

2 Milestone 3

2.1 Cleaning and Formatting Website Data

```
[57]: # Load Necessary Libraries
import numpy as np
import pandas as pd
import warnings
warnings.simplefilter(action='ignore', category=FutureWarning)
```

```
[59]: # Retrieving Data
player_site = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/6620/players/')
player_site
```

```
[59]: [          No.      Player Name \
0              2      Poulter
1              3      A. Skinner
2              4    Wong-Orantes
3              7      Carlini
4             10      Larson
5             11      Drews
6             12    Thompson
7             15    Washington
8             16      Rettke
9             22    Plummer
10            23      Cook
11            24    Ogbogu
12            NaN    Karch Kiraly
13    Reserve player (AP)    Reserve player (AP)
14              1      Hancock
15 *Reserve Player (AP) policy *Reserve Player (AP) policy

      Position
```

```

0          S
1         OH
2          L
3          S
4         OH
5          O
6          O
7         MB
8         MB
9         OH
10        OH
11        MB
12       COACH
13   Reserve player (AP)
14          S
15 *Reserve Player (AP) policy ]

```

```
[61]: #Determinen number of tables
print(f'Total tables: {len(player_site)}')
```

Total tables: 1

```
[63]: #Move HTML table into data frame
usa_players = player_site[0]
usa_players
```

```
[63]:
```

	No.	Player Name \
0	2	Poulter
1	3	A. Skinner
2	4	Wong-Orantes
3	7	Carlini
4	10	Larson
5	11	Drews
6	12	Thompson
7	15	Washington
8	16	Rettke
9	22	Plummer
10	23	Cook
11	24	Ogbogu
12	NaN	Karch Kiraly
13	Reserve player (AP)	Reserve player (AP)
14	1	Hancock
15	*Reserve Player (AP) policy	*Reserve Player (AP) policy

```

          Position
0          S
1         OH
2          L

```

```

3           S
4           OH
5           O
6           O
7           MB
8           MB
9           OH
10          OH
11          MB
12          COACH
13          Reserve player (AP)
14          S
15  *Reserve Player (AP) policy

```

```

[65]: #Step 1: Add column for team name
      usa_players.insert(0, 'Name', 'USA')
      usa_players

```

```

[65]:   Name           No.      Player Name \
0   USA             2        Poulter
1   USA             3         A. Skinner
2   USA             4      Wong-Orantes
3   USA             7        Carlini
4   USA            10        Larson
5   USA            11         Drews
6   USA            12       Thompson
7   USA            15      Washington
8   USA            16        Rettke
9   USA            22        Plummer
10  USA            23         Cook
11  USA            24        Ogbogu
12  USA            NaN      Karch Kiraly
13  USA      Reserve player (AP)  Reserve player (AP)
14  USA             1        Hancock
15  USA  *Reserve Player (AP) policy  *Reserve Player (AP) policy

```

```

          Position
0           S
1          OH
2           L
3           S
4          OH
5           O
6           O
7          MB
8          MB
9          OH

```

```

10                                OH
11                                MB
12                                COACH
13        Reserve player (AP)
14                                S
15    *Reserve Player (AP) policy

```

```

[67]: #Step 2: Remove coach and reserve players
index = usa_players[usa_players['Position'] == 'COACH'].index[0]
usa_filtered = usa_players.loc[:index-1]
usa_filtered

```

```

[67]:   Name No.   Player Name Position
0   USA    2     Poulter         S
1   USA    3   A. Skinner        OH
2   USA    4 Wong-Orantes        L
3   USA    7     Carlini         S
4   USA   10     Larson          OH
5   USA   11     Drews           O
6   USA   12     Thompson        O
7   USA   15   Washington        MB
8   USA   16     Rettke          MB
9   USA   22     Plummer         OH
10  USA   23     Cook            OH
11  USA   24     Ogbogu          MB

```

```

[69]: #Step 3: Replace column names
usa_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
usa_filtered

```

```

[69]:   Team JerseyNumber   Name Position
0   USA           2     Poulter         S
1   USA           3   A. Skinner        OH
2   USA           4 Wong-Orantes        L
3   USA           7     Carlini         S
4   USA          10     Larson          OH
5   USA          11     Drews           O
6   USA          12     Thompson        O
7   USA          15   Washington        MB
8   USA          16     Rettke          MB
9   USA          22     Plummer         OH
10  USA          23     Cook            OH
11  USA          24     Ogbogu          MB

```

```

[71]: #Step 4: Replace position acronyms with full name
def player_position(x):
    if any('L' in str(item) for item in x):

```

```

        return 'Libero'
    elif any('S' in str(item) for item in x):
        return 'Setter'
    elif any('MB' for item in x):
        return 'Middle-Blocker'
    elif any('OH' in str(item) for item in x):
        return 'Outside Hitter'
    elif any('O' in str(item) for item in x):
        return 'Opposite'
    else:
        return ['Position']

```

```

usa_filtered['Position'] = usa_filtered['Position'].apply(player_position)
usa_filtered

```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\2031534157.py:16:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```

    usa_filtered['Position'] = usa_filtered['Position'].apply(player_position)

```

```

[71]:
   Team JerseyNumber      Name      Position
0   USA           2    Poulter         Setter
1   USA           3  A. Skinner  Middle-Blocker
2   USA           4 Wong-Orantes         Libero
3   USA           7    Carlini         Setter
4   USA          10    Larson  Middle-Blocker
5   USA          11    Drews  Middle-Blocker
6   USA          12    Thompson  Middle-Blocker
7   USA          15 Washington  Middle-Blocker
8   USA          16    Rettke  Middle-Blocker
9   USA          22    Plummer  Middle-Blocker
10  USA          23     Cook  Middle-Blocker
11  USA          24    Ogbogu  Middle-Blocker

```

```

[73]: # Repeat for other teams
player_site1 = pd.read_html('https://en.volleyballworld.com/volleyball/
    competitions/volleyball-olympic-games-paris-2024/teams/women/6619/players/')
brazil_players = player_site1[0]
brazil_players.insert(0, 'Name', 'BRA')
index1 = brazil_players[brazil_players['Position'] == 'COACH'].index[0]
brazil_filtered = brazil_players.loc[:index1-1]
brazil_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
brazil_filtered['Position'] = brazil_filtered['Position'].apply(player_position)

```

```

player_site2 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/7028/players/')
china_players = player_site2[0]
china_players.insert(0, 'Name', 'CHI')
index2 = china_players[china_players['Position'] == 'COACH'].index[0]
china_filtered = china_players.loc[:index2-1]
china_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
china_filtered['Position'] = china_filtered['Position'].apply(player_position)

player_site3 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/6616/players/')
domrep_players = player_site3[0]
domrep_players.insert(0, 'Name', 'DOM')
index3 = domrep_players[domrep_players['Position'] == 'COACH'].index[0]
domrep_filtered = domrep_players.loc[:index3-1]
domrep_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
domrep_filtered['Position'] = domrep_filtered['Position'].apply(player_position)

player_site4 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/6615/players/')
france_players = player_site4[0]
france_players.insert(0, 'Name', 'FRA')
index4 = france_players[france_players['Position'] == 'COACH'].index[0]
france_filtered = france_players.loc[:index4-1]
france_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
france_filtered['Position'] = france_filtered['Position'].apply(player_position)

player_site5 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/7024/players/')
italy_players = player_site5[0]
italy_players.insert(0, 'Name', 'ITA')
index5 = italy_players[italy_players['Position'] == 'COACH'].index[0]
italy_filtered = italy_players.loc[:index5-1]
italy_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
italy_filtered['Position'] = italy_filtered['Position'].apply(player_position)

player_site6 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/7026/players/')
japan_players = player_site6[0]
japan_players.insert(0, 'Name', 'JAP')
index6 = japan_players[japan_players['Position'] == 'COACH'].index[0]
japan_filtered = japan_players.loc[:index6-1]
japan_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
japan_filtered['Position'] = japan_filtered['Position'].apply(player_position)

player_site7 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/7027/players/')

```

```

kenya_players = player_site7[0]
kenya_players.insert(0, 'Name', 'KEN')
index7 = kenya_players[kenya_players['Position'] == 'COACH'].index[0]
kenya_filtered = kenya_players.loc[:index7-1]
kenya_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
kenya_filtered['Position'] = kenya_filtered['Position'].apply(player_position)

player_site8 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/7023/players/')
neth_players = player_site8[0]
neth_players.insert(0, 'Name', 'NET')
index8 = neth_players[neth_players['Position'] == 'COACH'].index[0]
neth_filtered = neth_players.loc[:index8-1]
neth_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
neth_filtered['Position'] = neth_filtered['Position'].apply(player_position)

player_site9 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/6612/players/')
poland_players = player_site9[0]
poland_players.insert(0, 'Name', 'POL')
index9 = poland_players[poland_players['Position'] == 'COACH'].index[0]
poland_filtered = poland_players.loc[:index9-1]
poland_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
poland_filtered['Position'] = poland_filtered['Position'].apply(player_position)

player_site10 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/6617/players/')
serbia_players = player_site10[0]
serbia_players.insert(0, 'Name', 'SER')
index10 = serbia_players[serbia_players['Position'] == 'COACH'].index[0]
serbia_filtered = serbia_players.loc[:index10-1]
serbia_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
serbia_filtered['Position'] = serbia_filtered['Position'].apply(player_position)

player_site11 = pd.read_html('https://en.volleyballworld.com/volleyball/
↳competitions/volleyball-olympic-games-paris-2024/teams/women/6618/players/')
turkey_players = player_site11[0]
turkey_players.insert(0, 'Name', 'TUR')
index11 = turkey_players[turkey_players['Position'] == 'COACH'].index[0]
turkey_filtered = turkey_players.loc[:index11-1]
turkey_filtered.columns = ['Team', 'JerseyNumber', 'Name', 'Position']
turkey_filtered['Position'] = turkey_filtered['Position'].apply(player_position)

```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:8:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
brazil_filtered['Position'] =  
brazil_filtered['Position'].apply(player_position)  
C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:16:  
SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
china_filtered['Position'] = china_filtered['Position'].apply(player_position)  
C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:24:  
SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
domrep_filtered['Position'] =  
domrep_filtered['Position'].apply(player_position)  
C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:32:  
SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
france_filtered['Position'] =  
france_filtered['Position'].apply(player_position)  
C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:40:  
SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
italy_filtered['Position'] = italy_filtered['Position'].apply(player_position)  
C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:48:  
SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row_indexer,col_indexer] = value instead
```

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
japan_filtered['Position'] = japan_filtered['Position'].apply(player_position)  
C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:56:
```

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
kenya_filtered['Position'] = kenya_filtered['Position'].apply(player_position)
```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:64:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
neth_filtered['Position'] = neth_filtered['Position'].apply(player_position)
```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:72:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
poland_filtered['Position'] =  
poland_filtered['Position'].apply(player_position)
```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:80:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
serbia_filtered['Position'] =  
serbia_filtered['Position'].apply(player_position)
```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\1905239839.py:88:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
turkey_filtered['Position'] =  
turkey_filtered['Position'].apply(player_position)
```

```
[75]: # Step 5: Merge teams together  
olympic_teams = pd.concat([usa_filtered, brazil_filtered, china_filtered,   
    domrep_filtered, france_filtered,
```

```

        italy_filtered, japan_filtered, kenya_filtered, neth_filtered,
        ↪ poland_filtered,
        serbia_filtered, turkey_filtered], join='outer')

```

```

[77]: # Step 6: Add full team name
def team_name(x):
    if x == 'USA':
        return 'USA'
    elif x == 'BRA':
        return 'Brazil'
    elif x == 'DOM':
        return 'Dominican Republic'
    elif x == 'FRA':
        return 'France'
    elif x == 'CHI':
        return 'China'
    elif x == 'ITA':
        return 'Italy'
    elif x == 'JAP':
        return 'Japan'
    elif x == 'KEN':
        return 'Kenya'
    elif x == 'NET':
        return 'Netherlands'
    elif x == 'POL':
        return 'Poland'
    elif x == 'SER':
        return 'Serbia'
    elif x == 'TUR':
        return 'Turkey'
    else:
        return ['Team']

olympic_teams['TeamName'] = olympic_teams['Team'].apply(team_name)
olympic_teams

```

```

[77]:

```

	Team	JerseyNumber	Name	Position	TeamName
0	USA	2	Poulter	Setter	USA
1	USA	3	A. Skinner	Middle-Blocker	USA
2	USA	4	Wong-Orantes	Libero	USA
3	USA	7	Carlini	Setter	USA
4	USA	10	Larson	Middle-Blocker	USA
..	...	...	...	...	...
7	TUR	14	Erdem	Middle-Blocker	Turkey
8	TUR	18	Güneş	Middle-Blocker	Turkey
9	TUR	19	Kalaç	Middle-Blocker	Turkey
10	TUR	22	Aydın	Middle-Blocker	Turkey

11 TUR 99 Karakurt Middle-Blocker Turkey

[157 rows x 5 columns]

2.2 Ethical Implications

The data was retrieve from a website that is public and based in the US. The data was reviewed as accurate compared to knowledge of the US Olympic Volleyball team members.

Changes made to this web data set included adding team name based on the website used to retrieve names, removing the coach and any alternate players, renamed columns, replaced player position initials with full name, merged the team data into one data frame, added the full team name rather than just the abbreviations.

There are no ethical implications with the modifications to this data set. It was truly cleaned and transformed to make it easier to use for the final steps. Fields were added based on facts, modified to remove abiguity (removed abbreviations).

The About page of Volleyball World (<https://en.volleyballworld.com/about-us/about-volleyball-world>) indicates they starts in 2021 with key partners and have continued to growth since.

3 Milestone 4

3.1 Cleaning/Formatting API Source

```
[82]: import requests
      from bs4 import BeautifulSoup
      import json
      import pandas as pd
      import urllib.request, urllib.parse
      from urllib.error import HTTPError, URLError
      import warnings
      warnings.simplefilter(action='ignore', category=FutureWarning)

[84]: # Connect to an API of your choice and do a simple data pull
      api_results = 'https://www.thesportsdb.com/api/v1/json/3/eventsseason.php?
                  ↪id=5411&s=2024'

[86]: # Connect to the API and do a "Get" call/operation on the API to return the API
      ↪data
      oly_results = requests.get(api_results)
      olympic_df = oly_results.json()
      # Put the API results into a dataframe
      olympic_pretty = pd.json_normalize(olympic_df['events'], meta=['idEvent',
                  ↪'idAPIfootball', 'strEvent', 'strEventAlternate', 'strFilename', 'strSport',
                  ↪'idLeague',
      'strLeague', 'strLeagueBadge', 'strSeason', 'strDescriptionEN', 'strHomeTeam',
                  ↪'strAwayTeam',
```

```

'intHomeScore', 'intRound', 'intAwayScore', 'intSpectators', 'strOfficial',␣
↳'strTimestamp',
'dateEvent', 'dateEventLocal', 'strTime', 'strTimeLocal', 'strGroup',␣
↳'idHomeTeam', 'strHomeTeamBadge',
'idAwayTeam', 'strAwayTeamBadge', 'intScore', 'intScoreVotes', 'strResult',
'idVenue', 'strVenue', 'strCountry', 'strCity', 'strPoster', 'strSquare',␣
↳'strFanart', 'strThumb',
'strBanner', 'strMap', 'strTweet1', 'strTweet2', 'strTweet3', 'strVideo',␣
↳'strStatus',
'strPostponed', 'strLocked'])
olympic_pretty

```

```

[86]:      idEvent idAPIfootball      strEvent \
0   2078495      166140 Italy Volleyball Women vs Dominican Republic V...
1   2078496      166141 Poland Volleyball Women vs Japan Volleyball Women
2   2078497      166142 Turkey Volleyball Women vs Netherlands Volleyb...
3   2078498      166144      USA Volleyball Women vs China Volleyball Women
4   2078500      166146 France Volleyball Women vs Serbia Volleyball W...
5   2078499      166145      USA Volleyball Women vs Serbia Volleyball Women
6   2078501      166148 Turkey Volleyball Women vs Dominican Republic ...
7   2078502      166149 Brazil Volleyball Women vs Japan Volleyball Women
8   2078503      166150 Italy Volleyball Women vs Netherlands Volleyba...
9   2078504      166151 France Volleyball Women vs China Volleyball Women
10  2078505      166152 Netherlands Volleyball Women vs Dominican Repu...
11  2078506      166154 Italy Volleyball Women vs Turkey Volleyball Women
12  2078507      166155      France Volleyball Women vs USA Volleyball Women
13  2078508      166156 China Volleyball Women vs Serbia Volleyball Women
14  2078509      166157 Brazil Volleyball Women vs Poland Volleyball W...
15  2135632      169814 China Volleyball Women vs Turkey Volleyball Women
16  2135633      169815 Brazil Volleyball Women vs Dominican Republic ...
17  2135634      169816      USA Volleyball Women vs Poland Volleyball Women
18  2135635      169817 Italy Volleyball Women vs Serbia Volleyball Women
19  2135940      169822      Brazil Volleyball Women vs USA Volleyball Women
20  2135941      169823 Turkey Volleyball Women vs Italy Volleyball Women
21  2138445      170202 Brazil Volleyball Women vs Turkey Volleyball W...
22  2138446      170203      USA Volleyball Women vs Italy Volleyball Women

                                strEventAlternate \
0   Dominican Republic Volleyball Women @ Italy Vo...
1   Japan Volleyball Women @ Poland Volleyball Women
2   Netherlands Volleyball Women @ Turkey Volleyba...
3   China Volleyball Women @ USA Volleyball Women
4   Serbia Volleyball Women @ France Volleyball Women
5   Serbia Volleyball Women @ USA Volleyball Women
6   Dominican Republic Volleyball Women @ Turkey V...
7   Japan Volleyball Women @ Brazil Volleyball Women
8   Netherlands Volleyball Women @ Italy Volleybal...

```

9 China Volleyball Women @ France Volleyball Women
 10 Dominican Republic Volleyball Women @ Netherla...
 11 Turkey Volleyball Women @ Italy Volleyball Women
 12 USA Volleyball Women @ France Volleyball Women
 13 Serbia Volleyball Women @ China Volleyball Women
 14 Poland Volleyball Women @ Brazil Volleyball Women
 15 Turkey Volleyball Women @ China Volleyball Women
 16 Dominican Republic Volleyball Women @ Brazil V...
 17 Poland Volleyball Women @ USA Volleyball Women
 18 Serbia Volleyball Women @ Italy Volleyball Women
 19 USA Volleyball Women @ Brazil Volleyball Women
 20 Italy Volleyball Women @ Turkey Volleyball Women
 21 Turkey Volleyball Women @ Brazil Volleyball Women
 22 Italy Volleyball Women @ USA Volleyball Women

					strFilename	strSport	idLeague	\
0	Olympics	Volleyball	Women	2024-07-28	Italy Vol...	Volleyball	5411	
1	Olympics	Volleyball	Women	2024-07-28	Poland Vo...	Volleyball	5411	
2	Olympics	Volleyball	Women	2024-07-29	Turkey Vo...	Volleyball	5411	
3	Olympics	Volleyball	Women	2024-07-29	USA Volle...	Volleyball	5411	
4	Olympics	Volleyball	Women	2024-07-29	France Vo...	Volleyball	5411	
5	Olympics	Volleyball	Women	2024-07-31	USA Volle...	Volleyball	5411	
6	Olympics	Volleyball	Women	2024-08-01	Turkey Vo...	Volleyball	5411	
7	Olympics	Volleyball	Women	2024-08-01	Brazil Vo...	Volleyball	5411	
8	Olympics	Volleyball	Women	2024-08-01	Italy Vol...	Volleyball	5411	
9	Olympics	Volleyball	Women	2024-08-01	France Vo...	Volleyball	5411	
10	Olympics	Volleyball	Women	2024-08-03	Netherlan...	Volleyball	5411	
11	Olympics	Volleyball	Women	2024-08-04	Italy Vol...	Volleyball	5411	
12	Olympics	Volleyball	Women	2024-08-04	France Vo...	Volleyball	5411	
13	Olympics	Volleyball	Women	2024-08-04	China Vol...	Volleyball	5411	
14	Olympics	Volleyball	Women	2024-08-04	Brazil Vo...	Volleyball	5411	
15	Olympics	Volleyball	Women	2024-08-06	China Vol...	Volleyball	5411	
16	Olympics	Volleyball	Women	2024-08-06	Brazil Vo...	Volleyball	5411	
17	Olympics	Volleyball	Women	2024-08-06	USA Volle...	Volleyball	5411	
18	Olympics	Volleyball	Women	2024-08-06	Italy Vol...	Volleyball	5411	
19	Olympics	Volleyball	Women	2024-08-08	Brazil Vo...	Volleyball	5411	
20	Olympics	Volleyball	Women	2024-08-08	Turkey Vo...	Volleyball	5411	
21	Olympics	Volleyball	Women	2024-08-10	Brazil Vo...	Volleyball	5411	
22	Olympics	Volleyball	Women	2024-08-11	USA Volle...	Volleyball	5411	

	strLeague	\
0	Olympics Volleyball Women	
1	Olympics Volleyball Women	
2	Olympics Volleyball Women	
3	Olympics Volleyball Women	
4	Olympics Volleyball Women	
5	Olympics Volleyball Women	

6 Olympics Volleyball Women
7 Olympics Volleyball Women
8 Olympics Volleyball Women
9 Olympics Volleyball Women
10 Olympics Volleyball Women
11 Olympics Volleyball Women
12 Olympics Volleyball Women
13 Olympics Volleyball Women
14 Olympics Volleyball Women
15 Olympics Volleyball Women
16 Olympics Volleyball Women
17 Olympics Volleyball Women
18 Olympics Volleyball Women
19 Olympics Volleyball Women
20 Olympics Volleyball Women
21 Olympics Volleyball Women
22 Olympics Volleyball Women

	strLeagueBadge	strSeason	\
0	https://www.thesportsdb.com/images/media/leagu...	2024	
1	https://www.thesportsdb.com/images/media/leagu...	2024	
2	https://www.thesportsdb.com/images/media/leagu...	2024	
3	https://www.thesportsdb.com/images/media/leagu...	2024	
4	https://www.thesportsdb.com/images/media/leagu...	2024	
5	https://www.thesportsdb.com/images/media/leagu...	2024	
6	https://www.thesportsdb.com/images/media/leagu...	2024	
7	https://www.thesportsdb.com/images/media/leagu...	2024	
8	https://www.thesportsdb.com/images/media/leagu...	2024	
9	https://www.thesportsdb.com/images/media/leagu...	2024	
10	https://www.thesportsdb.com/images/media/leagu...	2024	
11	https://www.thesportsdb.com/images/media/leagu...	2024	
12	https://www.thesportsdb.com/images/media/leagu...	2024	
13	https://www.thesportsdb.com/images/media/leagu...	2024	
14	https://www.thesportsdb.com/images/media/leagu...	2024	
15	https://www.thesportsdb.com/images/media/leagu...	2024	
16	https://www.thesportsdb.com/images/media/leagu...	2024	
17	https://www.thesportsdb.com/images/media/leagu...	2024	
18	https://www.thesportsdb.com/images/media/leagu...	2024	
19	https://www.thesportsdb.com/images/media/leagu...	2024	
20	https://www.thesportsdb.com/images/media/leagu...	2024	
21	https://www.thesportsdb.com/images/media/leagu...	2024	
22	https://www.thesportsdb.com/images/media/leagu...	2024	

	strDescriptionEN	strHomeTeam	\
0		Italy Volleyball Women	
1		Poland Volleyball Women	
2		Turkey Volleyball Women	

3	USA Volleyball Women
4	France Volleyball Women
5	USA Volleyball Women
6	Turkey Volleyball Women
7	Brazil Volleyball Women
8	Italy Volleyball Women
9	France Volleyball Women
10	Netherlands Volleyball Women
11	Italy Volleyball Women
12	France Volleyball Women
13	China Volleyball Women
14	Brazil Volleyball Women
15	China Volleyball Women
16	Brazil Volleyball Women
17	USA Volleyball Women
18	Italy Volleyball Women
19	Brazil Volleyball Women
20	Turkey Volleyball Women
21	Brazil Volleyball Women
22	USA Volleyball Women

	strAwayTeam	intHomeScore	intRound	intAwayScore	\
0	Dominican Republic Volleyball Women	3	1	1	
1	Japan Volleyball Women	3	1	1	
2	Netherlands Volleyball Women	3	1	2	
3	China Volleyball Women	2	1	3	
4	Serbia Volleyball Women	0	1	3	
5	Serbia Volleyball Women	3	2	2	
6	Dominican Republic Volleyball Women	3	2	1	
7	Japan Volleyball Women	3	2	0	
8	Netherlands Volleyball Women	3	2	0	
9	China Volleyball Women	0	2	3	
10	Dominican Republic Volleyball Women	1	3	3	
11	Turkey Volleyball Women	3	3	0	
12	USA Volleyball Women	0	3	3	
13	Serbia Volleyball Women	3	3	1	
14	Poland Volleyball Women	3	3	0	
15	Turkey Volleyball Women	2	0	3	
16	Dominican Republic Volleyball Women	3	0	0	
17	Poland Volleyball Women	3	0	0	
18	Serbia Volleyball Women	3	0	0	
19	USA Volleyball Women	2	0	3	
20	Italy Volleyball Women	0	0	3	
21	Turkey Volleyball Women	3	0	1	
22	Italy Volleyball Women	0	0	3	

intSpectators	strOfficial	strTimestamp	dateEvent	dateEventLocal	\
---------------	-------------	--------------	-----------	----------------	---

0	None	None	2024-07-28T07:00:00	2024-07-28	2024-07-28
1	None	None	2024-07-28T11:00:00	2024-07-28	2024-07-28
2	None	None	2024-07-29T07:00:00	2024-07-29	2024-07-29
3	None	None	2024-07-29T15:00:00	2024-07-29	2024-07-29
4	None	None	2024-07-29T19:00:00	2024-07-29	2024-07-29
5	None	None	2024-07-31T15:00:00	2024-07-31	2024-07-31
6	None	None	2024-08-01T07:00:00	2024-08-01	2024-08-01
7	None	None	2024-08-01T11:00:00	2024-08-01	2024-08-01
8	None	None	2024-08-01T15:00:00	2024-08-01	2024-08-01
9	None	None	2024-08-01T19:00:00	2024-08-01	2024-08-01
10	None	None	2024-08-03T07:00:00	2024-08-03	2024-08-03
11	None	None	2024-08-04T07:00:00	2024-08-04	2024-08-04
12	None	None	2024-08-04T11:00:00	2024-08-04	2024-08-04
13	None	None	2024-08-04T15:00:00	2024-08-04	2024-08-04
14	None	None	2024-08-04T19:00:00	2024-08-04	2024-08-04
15	None	None	2024-08-06T07:00:00	2024-08-06	2024-08-06
16	None	None	2024-08-06T11:00:00	2024-08-06	2024-08-06
17	None	None	2024-08-06T15:00:00	2024-08-06	2024-08-06
18	None	None	2024-08-06T19:00:00	2024-08-06	2024-08-06
19	None	None	2024-08-08T14:00:00	2024-08-08	2024-08-08
20	None	None	2024-08-08T18:00:00	2024-08-08	2024-08-08
21	None	None	2024-08-10T15:15:00	2024-08-10	2024-08-10
22	None	None	2024-08-11T11:00:00	2024-08-11	2024-08-11

	strTime	strTimeLocal	strGroup	idHomeTeam	\
0	07:00:00			142119	
1	11:00:00			142118	
2	07:00:00			142111	
3	15:00:00			142108	
4	19:00:00			148427	
5	15:00:00			142108	
6	07:00:00			142111	
7	11:00:00			142109	
8	15:00:00			142119	
9	19:00:00			148427	
10	07:00:00			142114	
11	07:00:00			142119	
12	11:00:00			148427	
13	15:00:00			142112	
14	19:00:00			142109	
15	07:00:00			142112	
16	11:00:00			142109	
17	15:00:00			142108	
18	19:00:00	21:00:00		142119	
19	14:00:00	16:00:00		142109	
20	18:00:00			142111	
21	15:15:00			142109	

22 11:00:00

142108

	strHomeTeamBadge	idAwayTeam \
0	https://r2.thesportsdb.com/images/media/team/b...	142113
1	https://r2.thesportsdb.com/images/media/team/b...	142110
2	https://r2.thesportsdb.com/images/media/team/b...	142114
3	https://r2.thesportsdb.com/images/media/team/b...	142112
4	https://r2.thesportsdb.com/images/media/team/b...	142120
5	https://r2.thesportsdb.com/images/media/team/b...	142120
6	https://r2.thesportsdb.com/images/media/team/b...	142113
7	https://r2.thesportsdb.com/images/media/team/b...	142110
8	https://r2.thesportsdb.com/images/media/team/b...	142114
9	https://r2.thesportsdb.com/images/media/team/b...	142112
10	https://r2.thesportsdb.com/images/media/team/b...	142113
11	https://r2.thesportsdb.com/images/media/team/b...	142111
12	https://r2.thesportsdb.com/images/media/team/b...	142108
13	https://r2.thesportsdb.com/images/media/team/b...	142120
14	https://r2.thesportsdb.com/images/media/team/b...	142118
15	https://r2.thesportsdb.com/images/media/team/b...	142111
16	https://r2.thesportsdb.com/images/media/team/b...	142113
17	https://r2.thesportsdb.com/images/media/team/b...	142118
18	https://r2.thesportsdb.com/images/media/team/b...	142120
19	https://r2.thesportsdb.com/images/media/team/b...	142108
20	https://r2.thesportsdb.com/images/media/team/b...	142119
21	https://r2.thesportsdb.com/images/media/team/b...	142111
22	https://r2.thesportsdb.com/images/media/team/b...	142119

	strAwayTeamBadge	intScore	intScoreVotes \
0	https://r2.thesportsdb.com/images/media/team/b...	None	None
1	https://r2.thesportsdb.com/images/media/team/b...	None	None
2	https://r2.thesportsdb.com/images/media/team/b...	None	None
3	https://r2.thesportsdb.com/images/media/team/b...	None	None
4	https://r2.thesportsdb.com/images/media/team/b...	None	None
5	https://r2.thesportsdb.com/images/media/team/b...	None	None
6	https://r2.thesportsdb.com/images/media/team/b...	None	None
7	https://r2.thesportsdb.com/images/media/team/b...	None	None
8	https://r2.thesportsdb.com/images/media/team/b...	None	None
9	https://r2.thesportsdb.com/images/media/team/b...	None	None
10	https://r2.thesportsdb.com/images/media/team/b...	None	None
11	https://r2.thesportsdb.com/images/media/team/b...	None	None
12	https://r2.thesportsdb.com/images/media/team/b...	None	None
13	https://r2.thesportsdb.com/images/media/team/b...	None	None
14	https://r2.thesportsdb.com/images/media/team/b...	None	None
15	https://r2.thesportsdb.com/images/media/team/b...	None	None
16	https://r2.thesportsdb.com/images/media/team/b...	None	None
17	https://r2.thesportsdb.com/images/media/team/b...	None	None
18	https://r2.thesportsdb.com/images/media/team/b...	None	None

19	https://r2.thesportsdb.com/images/media/team/b...	None	None
20	https://r2.thesportsdb.com/images/media/team/b...	None	None
21	https://r2.thesportsdb.com/images/media/team/b...	None	None
22	https://r2.thesportsdb.com/images/media/team/b...	None	None

		strResult	idVenue	\
0	Italy Volleyball Women First: 25 19 Se...		30945	
1	Poland Volleyball Women First: 20 25 S...		30945	
2	Turkey Volleyball Women First: 19 25 S...		30945	
3	USA Volleyball Women First: 20 25 Seco...		30945	
4	France Volleyball Women First: 17 25 S...		30945	
5	USA Volleyball Women First: 25 17 Seco...		30945	
6	Turkey Volleyball Women First: 21 25 S...		30945	
7	Brazil Volleyball Women First: 25 20 S...		30945	
8	Italy Volleyball Women First: 29 27 Se...		30945	
9	France Volleyball Women First: 18 25 S...		30945	
10	Netherlands Volleyball Women First: 25 22 ...		30945	
11	Italy Volleyball Women First: 25 14 Se...		30945	
12	France Volleyball Women First: 27 29 S...		30945	
13	China Volleyball Women First: 21 25 Se...		30945	
14	Brazil Volleyball Women First: 25 21 S...		30945	
15	China Volleyball Women First: 25 23 Se...		0	
16	Brazil Volleyball Women First: 25 22 S...		0	
17	USA Volleyball Women First: 25 22 Seco...		0	
18	Italy Volleyball Women First: 26 24 Se...		0	
19	Brazil Volleyball Women First: 23 25 S...		0	
20	Turkey Volleyball Women First: 22 25 S...		0	
21	Brazil Volleyball Women First: 25 21 S...		0	
22	USA Volleyball Women First: 18 25 Seco...		0	

	strVenue	strCountry	strCity	strPoster	\
0	South Paris Arena 1	Italy	Paris		
1	South Paris Arena 1	Poland	Paris		
2	South Paris Arena 1	Turkey	Paris		
3	South Paris Arena 1	United States	Paris		
4	South Paris Arena 1	France	Paris		
5	South Paris Arena 1	United States	Paris		
6	South Paris Arena 1	Turkey	Paris		
7	South Paris Arena 1	Brazil	Paris		
8	South Paris Arena 1	Italy	Paris		
9	South Paris Arena 1	France	Paris		
10	South Paris Arena 1	The Netherlands	Paris		
11	South Paris Arena 1	Italy	Paris		
12	South Paris Arena 1	France	Paris		
13	South Paris Arena 1	China	Paris		
14	South Paris Arena 1	Brazil	Paris		
15		China			

16	Brazil
17	United States
18	Italy
19	Brazil
20	Turkey
21	Brazil
22	United States

	strSquare	strFanart	\
0	https://r2.thesportsdb.com/images/media/event/...	None	
1	https://r2.thesportsdb.com/images/media/event/...	None	
2	https://r2.thesportsdb.com/images/media/event/...	None	
3	https://r2.thesportsdb.com/images/media/event/...	None	
4	https://r2.thesportsdb.com/images/media/event/...	None	
5	https://r2.thesportsdb.com/images/media/event/...	None	
6	https://r2.thesportsdb.com/images/media/event/...	None	
7	https://r2.thesportsdb.com/images/media/event/...	None	
8	https://r2.thesportsdb.com/images/media/event/...	None	
9	https://r2.thesportsdb.com/images/media/event/...	None	
10	https://r2.thesportsdb.com/images/media/event/...	None	
11	https://r2.thesportsdb.com/images/media/event/...	None	
12	https://r2.thesportsdb.com/images/media/event/...	None	
13	https://r2.thesportsdb.com/images/media/event/...	None	
14	https://r2.thesportsdb.com/images/media/event/...	None	
15		None	
16		None	
17		None	
18		None	
19		None	
20		None	
21		None	
22		None	

	strThumb	strBanner	strMap	\
0	https://r2.thesportsdb.com/images/media/event/...		None	
1	https://r2.thesportsdb.com/images/media/event/...		None	
2	https://r2.thesportsdb.com/images/media/event/...		None	
3	https://r2.thesportsdb.com/images/media/event/...		None	
4	https://r2.thesportsdb.com/images/media/event/...		None	
5	https://r2.thesportsdb.com/images/media/event/...		None	
6	https://r2.thesportsdb.com/images/media/event/...		None	
7	https://r2.thesportsdb.com/images/media/event/...		None	
8	https://r2.thesportsdb.com/images/media/event/...		None	
9	https://r2.thesportsdb.com/images/media/event/...		None	
10	https://r2.thesportsdb.com/images/media/event/...		None	
11	https://r2.thesportsdb.com/images/media/event/...		None	
12	https://r2.thesportsdb.com/images/media/event/...		None	

13	https://r2.thesportsdb.com/images/media/event/...	None
14	https://r2.thesportsdb.com/images/media/event/...	None
15		None
16		None
17		None
18		None
19		None
20		None
21		None
22		None

	strTweet1	strTweet2	strTweet3	strVideo	\
0					
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18				https://www.youtube.com/watch?v=3n0790E8qTI	
19				https://www.youtube.com/watch?v=fPROAPx3A-Y	
20					
21					
22					

	strStatus	strPostponed	strLocked
0	FT	no	unlocked
1	FT	no	unlocked
2	FT	no	unlocked
3	FT	no	unlocked
4	FT	no	unlocked
5	FT	no	unlocked
6	FT	no	unlocked
7	FT	no	unlocked
8	FT	no	unlocked
9	FT	no	unlocked

10	FT	no	unlocked
11	FT	no	unlocked
12	FT	no	unlocked
13	FT	no	unlocked
14	FT	no	unlocked
15	FT	no	unlocked
16	FT	no	unlocked
17	FT	no	unlocked
18	FT	no	unlocked
19	FT	no	unlocked
20	FT	no	unlocked
21	FT	no	unlocked
22	FT	no	unlocked

3.1.1 First Transformation

```
[89]: # First Transformation: Remove columns not needed
olympic_short = olympic_pretty[['idEvent', 'strSport', 'idLeague', 'strLeague',
    ↪ 'strSeason', 'strHomeTeam', 'strAwayTeam',
    'intHomeScore', 'intRound', 'intAwayScore', 'dateEvent', 'strTime',
    ↪ 'idHomeTeam', 'idAwayTeam']]
olympic_short.head()
```

```
[89]:   idEvent   strSport idLeague      strLeague strSeason \
0  2078495  Volleyball   5411  Olympics Volleyball Women    2024
1  2078496  Volleyball   5411  Olympics Volleyball Women    2024
2  2078497  Volleyball   5411  Olympics Volleyball Women    2024
3  2078498  Volleyball   5411  Olympics Volleyball Women    2024
4  2078500  Volleyball   5411  Olympics Volleyball Women    2024

      strHomeTeam      strAwayTeam intHomeScore \
0  Italy Volleyball Women  Dominican Republic Volleyball Women    3
1  Poland Volleyball Women                Japan Volleyball Women    3
2  Turkey Volleyball Women      Netherlands Volleyball Women    3
3    USA Volleyball Women                China Volleyball Women    2
4  France Volleyball Women      Serbia Volleyball Women    0

   intRound intAwayScore   dateEvent   strTime idHomeTeam idAwayTeam
0         1           1  2024-07-28  07:00:00    142119    142113
1         1           1  2024-07-28  11:00:00    142118    142110
2         1           2  2024-07-29  07:00:00    142111    142114
3         1           3  2024-07-29  15:00:00    142108    142112
4         1           3  2024-07-29  19:00:00    148427    142120
```

3.1.2 Second Transformation

```
[92]: # Remove "Volleyball Women" from home and away teams to get just country
olympic_short['strHomeTeam'] = olympic_short['strHomeTeam'].str.replace('␣
↳Volleyball Women', '')
olympic_short['strAwayTeam'] = olympic_short['strAwayTeam'].str.replace('␣
↳Volleyball Women', '')
olympic_short.head()
```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\825328184.py:2:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
olympic_short['strHomeTeam'] = olympic_short['strHomeTeam'].str.replace('
Volleyball Women', '')
```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\825328184.py:3:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
olympic_short['strAwayTeam'] = olympic_short['strAwayTeam'].str.replace('
Volleyball Women', '')
```

```
[92]:   idEvent   strSport idLeague   strLeague strSeason \
0  2078495  Volleyball   5411  Olympics Volleyball Women   2024
1  2078496  Volleyball   5411  Olympics Volleyball Women   2024
2  2078497  Volleyball   5411  Olympics Volleyball Women   2024
3  2078498  Volleyball   5411  Olympics Volleyball Women   2024
4  2078500  Volleyball   5411  Olympics Volleyball Women   2024

   strHomeTeam   strAwayTeam  intHomeScore  intRound  intAwayScore \
0        Italy  Dominican Republic         3         1           1
1        Poland             Japan         3         1           1
2        Turkey      Netherlands         3         1           2
3         USA             China         2         1           3
4        France             Serbia         0         1           3

   dateEvent   strTime  idHomeTeam  idAwayTeam
0  2024-07-28  07:00:00     142119     142113
1  2024-07-28  11:00:00     142118     142110
2  2024-07-29  07:00:00     142111     142114
3  2024-07-29  15:00:00     142108     142112
4  2024-07-29  19:00:00     148427     142120
```

3.1.3 Third Transformation

```
[94]: # Convert match scores to integers
olympic_short['intHomeScore'] = pd.to_numeric(olympic_short['intHomeScore'],
errors='coerce')
olympic_short['intAwayScore'] = pd.to_numeric(olympic_short['intAwayScore'],
errors='coerce')
olympic_short.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23 entries, 0 to 22
Data columns (total 14 columns):
#   Column          Non-Null Count  Dtype
---  -
0   idEvent          23 non-null    object
1   strSport         23 non-null    object
2   idLeague         23 non-null    object
3   strLeague        23 non-null    object
4   strSeason        23 non-null    object
5   strHomeTeam      23 non-null    object
6   strAwayTeam      23 non-null    object
7   intHomeScore     23 non-null    int64
8   intRound         23 non-null    object
9   intAwayScore     23 non-null    int64
10  dateEvent        23 non-null    object
11  strTime          23 non-null    object
12  idHomeTeam       23 non-null    object
13  idAwayTeam       23 non-null    object
dtypes: int64(2), object(12)
memory usage: 2.6+ KB
```

```
C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\780722748.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#returning-a-view-versus-a-copy
olympic_short['intHomeScore'] = pd.to_numeric(olympic_short['intHomeScore'],
errors='coerce')
```

```
C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\780722748.py:3:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user\_guide/indexing.html#returning-a-view-versus-a-copy
olympic_short['intAwayScore'] = pd.to_numeric(olympic_short['intAwayScore'],
errors='coerce')
```

3.1.4 Fourth Transformation

```
[97]: # Create column that indicates the winner
olympic_short.loc[olympic_short['intHomeScore'] == 3, 'winning_team'] =
    ↳olympic_short['strHomeTeam']
olympic_short.loc[olympic_short['intAwayScore'] == 3, 'winning_team'] =
    ↳olympic_short['strAwayTeam']
olympic_short.head()
```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\3660063049.py:2:

SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
olympic_short.loc[olympic_short['intHomeScore'] == 3, 'winning_team'] =
olympic_short['strHomeTeam']
```

```
[97]:
```

	idEvent	strSport	idLeague	strLeague	strSeason	\
0	2078495	Volleyball	5411	Olympics Volleyball Women	2024	
1	2078496	Volleyball	5411	Olympics Volleyball Women	2024	
2	2078497	Volleyball	5411	Olympics Volleyball Women	2024	
3	2078498	Volleyball	5411	Olympics Volleyball Women	2024	
4	2078500	Volleyball	5411	Olympics Volleyball Women	2024	

	strHomeTeam	strAwayTeam	intHomeScore	intRound	intAwayScore	\
0	Italy	Dominican Republic	3	1	1	
1	Poland	Japan	3	1	1	
2	Turkey	Netherlands	3	1	2	
3	USA	China	2	1	3	
4	France	Serbia	0	1	3	

	dateEvent	strTime	idHomeTeam	idAwayTeam	winning_team
0	2024-07-28	07:00:00	142119	142113	Italy
1	2024-07-28	11:00:00	142118	142110	Poland
2	2024-07-29	07:00:00	142111	142114	Turkey
3	2024-07-29	15:00:00	142108	142112	China
4	2024-07-29	19:00:00	148427	142120	Serbia

3.1.5 Fifth Transformation

```
[102]: # Reorder Columns
olympic_final = olympic_short[['idEvent', 'strHomeTeam', 'strAwayTeam',
    ↳'intHomeScore', 'intAwayScore', 'intRound', 'winning_team', 'dateEvent',
    ↳'strSport', 'idLeague', 'strLeague', 'strSeason', 'strTime', 'idHomeTeam',
    ↳'idAwayTeam']]
olympic_final.head()
```

```
[102]:
```

	idEvent	strHomeTeam	strAwayTeam	intHomeScore	intAwayScore	\
0	2078495	Italy	Dominican Republic	3	1	
1	2078496	Poland	Japan	3	1	
2	2078497	Turkey	Netherlands	3	2	
3	2078498	USA	China	2	3	
4	2078500	France	Serbia	0	3	

	intRound	winning_team	dateEvent	strSport	idLeague	\
0	1	Italy	2024-07-28	Volleyball	5411	
1	1	Poland	2024-07-28	Volleyball	5411	
2	1	Turkey	2024-07-29	Volleyball	5411	
3	1	China	2024-07-29	Volleyball	5411	
4	1	Serbia	2024-07-29	Volleyball	5411	

	strLeague	strSeason	strTime	idHomeTeam	idAwayTeam
0	Olympics Volleyball Women	2024	07:00:00	142119	142113
1	Olympics Volleyball Women	2024	11:00:00	142118	142110
2	Olympics Volleyball Women	2024	07:00:00	142111	142114
3	Olympics Volleyball Women	2024	15:00:00	142108	142112
4	Olympics Volleyball Women	2024	19:00:00	148427	142120

```
[104]: # I tried to pull in more data from these other APIs but they didn't work. I
        ↳ beleive I need to have a paid account.
        # At this time I don't think they are neccessary to my ananylsis but will get
        ↳ it if needed for the final milestone piece.
        # Event Statistics by Id /api/v1/json/3/lookupeventstats.php?id=2078495
        # Event Lineup by Id /api/v1/json/3/lookuplineup.php?id=2078495
```

3.2 Ethical Implications

The API pulls data from a website that also has a web page for each item (<https://www.thesportsdb.com/season/5411-olympics-volleyball-women/2024>). This allows you to validate what has been pulled. Additional review was compared to other public information on the 2024 Olympic women's volleyball results.

Modifications to the data included putting the JSON into a pandas dataframe and made “pretty”. From there columns not needed were removed, columns modified to remove redundant text, format of columns changed from string to integer, created a column based on the sets won to list the winner, and reordered the columns. This was a small straightforward data set so minimal changes were needed .

The data was adjusted and new data created. Minimal adjustments to the retrieved data were made. The added field is based on other data received. Based on this there are no ethical implications with the modifications of this data set.

The Amout page of theSportsDb indicates they started in 2015 and have grown ever since based on the support of their users. It claims to be “one of the largest free sports databased on the net”, with user driven feature updates and changes. Supporters (paid suscribers) are accepted to keep them functioning, with supporters having more access than free users.e.

4 Milestone 5

4.1 Merging the Data and Storing in a Database

```
[109]: import sqlite3
with sqlite3.connect("DSC540db.db") as conn:
    cursor = conn.cursor()
```

```
[111]: # Import players table
player_reorder.to_sql('player_reorder', conn, if_exists='replace', index=False)
```

```
[111]: 76573
```

```
[113]: # Import team rosters
olympic_teams.to_sql('team_rosters', conn, if_exists='replace', index=False)
```

```
[113]: 157
```

```
[115]: # Import team results
olympic_final.to_sql('team_results', conn, if_exists='replace', index=False)
```

```
[115]: 23
```

```
[117]: # Removing players not in a nationality that played for the olympics
cursor.execute("""
    DELETE From player_reorder
    Where Nationality Not In ('Albania, Italy', 'Albania, USA', 'American_
↵ Samoa, USA', 'Andorra, Brazil', 'Argentina, Italy', 'Argentina, Poland',
'Argentina, USA', 'Australia, Italy', 'Australia, Netherlands', 'Australia,
↵ Poland', 'Australia, Serbia', 'Australia, USA',
'Austria, France', 'Austria, Serbia', 'Belarus, Poland', 'Belgium, France',
↵ 'Belgium, Poland', 'Belgium, Serbia',
'Bosnia and Herzegovina, Serbia', 'Brazil', 'Brazil, Chile', 'Brazil, Germany',
↵ 'Brazil, Israel', 'Brazil, Italy',
'Brazil, Luxembourg', 'Brazil, Mexico', 'Brazil, Portugal', 'Brazil, Rwanda',
↵ 'Brazil, Slovakia', 'Brazil, Spain',
'Brazil, Sweden', 'Brazil, Switzerland', 'Brazil, Turkey', 'Brazil, USA',
↵ 'Bulgaria, Italy', 'Bulgaria, USA',
'Canada, China', 'Canada, Italy', 'Canada, Poland', 'Canada, Serbia', 'Canada,
↵ USA', 'Chile, Italy', 'Chile, USA',
'China', 'China, Austria', 'China, France', 'China, Macau', 'China,
↵ Switzerland', 'China, USA', 'Costa Rica, Italy',
'Costa Rica, USA', 'Croatia, France', 'Croatia, Italy', 'Croatia, USA', 'Cuba,
↵ Italy', 'Cuba, USA', 'Cyprus, Brazil',
'Czech Republic, France', 'Czech Republic, Turkey', 'Dominican Republic',
↵ 'Dominican Republic, China',
```

'Dominican Republic, Puerto Rico', 'Dominican Republic, USA', 'Egypt, France',
 ↪ 'France', 'France, Albania',
 'France, Argentina', 'France, Benin', 'France, Brazil', 'France, Bulgaria',
 ↪ 'France, Cameroon', 'France, Canada',
 'France, Croatia', 'France, Denmark', 'France, French Polynesia', 'France',
 ↪ 'Georgia', 'France, Germany',
 'France, Greece', 'France, Guadeloupe', 'France, Morocco', 'France, New
 ↪ 'Caledonia', 'France, Norway',
 'France, Poland', 'France, Portugal', 'France, Reunion', 'France, Romania',
 ↪ 'France, Serbia', 'France, Slovakia',
 'France, Switzerland', 'France, Tunisia', 'France, USA', 'France, Wallis and
 ↪ 'Futuna', 'French Polynesia, France',
 'Georgia, Brazil', 'Germany, Brazil', 'Germany, Italy', 'Germany, Netherlands',
 ↪ 'Germany, Poland', 'Germany, Serbia',
 'Germany, Turkey', 'Germany, USA', 'Greece, Poland', 'Greece, Serbia', 'Greece',
 ↪ 'USA', 'Guadeloupe, France', 'Guam, USA',
 'Hungary, Italy', 'Hungary, Serbia', 'Hungary, USA', 'Ireland, USA', 'Israel',
 ↪ 'Brazil', 'Israel, USA', 'Italy', 'Italy, Albania',
 'Italy, Argentina', 'Italy, Belgium', 'Italy, Brazil', 'Italy, Bulgaria',
 ↪ 'Italy, Croatia', 'Italy, Cuba',
 'Italy, Dominican Republic', 'Italy, France', 'Italy, Germany', 'Italy',
 ↪ 'Greece', 'Italy, Hungary', 'Italy, Iceland',
 'Italy, Italy', 'Italy, Lithuania', 'Italy, Luxembourg', 'Italy, Moldova',
 ↪ 'Republic of', 'Italy, Morocco', 'Italy, Nigeria',
 'Italy, Peru', 'Italy, Poland', 'Italy, Romania', 'Italy, Russia', 'Italy, San
 ↪ 'Marino', 'Italy, Serbia', 'Italy, Slovenia',
 'Italy, South Korea', 'Italy, South Sudan', 'Italy, Spain', 'Italy',
 ↪ 'Switzerland', 'Italy, United Kingdom', 'Italy, USA',
 'Jamaica, USA', 'Japan', 'Japan, Canada', 'Japan, China', 'Japan, Cuba',
 ↪ 'Japan, Ghana', 'Japan, South Korea', 'Japan, USA',
 'Kazakhstan, Turkey', 'Latvia, France', 'Lithuania, USA', 'Luxembourg, Italy',
 ↪ 'Malta, Poland', 'Mauritius, France', 'Mexico, USA',
 'Mongolia, USA', 'Montenegro, Serbia', 'Morocco, France', 'Morocco, Italy',
 ↪ 'Netherlands', 'Netherlands Antilles',
 'Netherlands, Argentina', 'Netherlands, Australia', 'Netherlands, Germany',
 ↪ 'Netherlands, Italy', 'Netherlands, Luxembourg',
 'Netherlands, Netherlands', 'Netherlands, Poland', 'Netherlands, Russia',
 ↪ 'Netherlands, USA', 'New Zealand, Italy',
 'Nigeria, Japan', 'North Macedonia, Serbia', 'Norway, Italy', 'Norway, Poland',
 ↪ 'Panama, USA', 'Peru, Italy', 'Peru, Japan',
 'Philippines, Italy', 'Philippines, Japan', 'Philippines, USA', 'Poland',
 ↪ 'Poland, Canada', 'Poland, France', 'Poland, Germany',
 'Poland, Greece', 'Poland, Norway', 'Poland, Ukraine', 'Poland, United
 ↪ 'Kingdom', 'Poland, USA', 'Portugal, Brazil', 'Puerto Rico, USA',

```
'Romania, Turkey', 'Russia, Italy', 'Russia, Poland', 'Russia, Serbia',
↳ 'Russia, Turkey', 'Russia, USA', 'San Marino, Italy',
'Scotland, Poland', 'Serbia', 'Serbia, Bosnia and Herzegovina', 'Serbia,
↳ Croatia', 'Serbia, Hungary', 'Serbia, Italy',
'Serbia, Montenegro', 'Serbia, Oman', 'Serbia, Slovakia', 'Serbia, Spain',
↳ 'Serbia, Sweden', 'Serbia, Ukraine',
'South Korea, China', 'South Korea, USA', 'Spain, Italy', 'Sweden,
↳ Netherlands', 'Sweden, Turkey', 'Sweden, USA',
'Switzerland, Brazil', 'Switzerland, Italy', 'Switzerland, Netherlands',
↳ 'Switzerland, Poland', 'Switzerland, Serbia',
'Switzerland, Turkey', 'Thailand, USA', 'Trinidad and Tobago, Turkey',
↳ 'Turkey', 'Turkey, Albania', 'Turkey, Algeria',
'Turkey, Belarus', 'Turkey, Bosnia and Herzegovina', 'Turkey, Bulgaria',
↳ 'Turkey, Canada', 'Turkey, Cuba', 'Turkey, Estonia',
'Turkey, France', 'Turkey, Germany', 'Turkey, Kosovo', 'Turkey, Latvia',
↳ 'Turkey, Netherlands', 'Turkey, Russia', 'Turkey, Serbia',
'Turkey, United Kingdom', 'Turkey, United States Minor Outlying Islands',
↳ 'Turkey, USA', 'Ukraine, Turkey', 'United Kingdom, Italy',
'United Kingdom, Japan', 'Uruguay, Italy', 'USA', 'USA, American Samoa', 'USA,
↳ Argentina', 'USA, Asia/Pacific Region', 'USA, Australia',
'USA, Brazil', 'USA, Cameroon', 'USA, Canada', 'USA, Chile', 'USA, China',
↳ 'USA, Croatia', 'USA, Egypt', 'USA, El Salvador', 'USA, Germany',
'USA, Hungary', 'USA, Ireland', 'USA, Israel', 'USA, Italy', 'USA, Japan',
↳ 'USA, Lebanon', 'USA, Liechtenstein', 'USA, Luxembourg',
'USA, Mexico', 'USA, Morocco', 'USA, Netherlands', 'USA, New Zealand', 'USA,
↳ Peru', 'USA, Philippines', 'USA, Poland', 'USA, Puerto Rico',
'USA, Romania', 'USA, Serbia', 'USA, South Africa', 'USA, Ukraine', 'USA,
↳ United Kingdom', 'USA, Virgin Islands, U.S.', 'Virgin Islands, U.S.',
'Virgin Islands, U.S., USA')
;""")
```

```
# I tried this, but it didn't work for some reason. It would in normal SQL.
# Removing players not in a nationality that played for the olympics
#cursor.execute("""
#     DELETE From player_reorder
#     Where Nationality Not In ('%Brazil%', '%China%', '%Dominican
↳ Republic%', '%France%', '%Italy%', '%Japan%', '%Netherlands%',
#     '%Poland%', '%Serbia%', '%Turkey%', '%USA%', '%U.S.%')
#     ;""")
```

[117]: <sqlite3.Cursor at 0x270710d1ac0>

```
[119]: # Split Season
cursor.execute("""
        CREATE TABLE player_new AS
```

```

        Select *, TRIM(substr(Season,1,7 )) as StartSeason,
        ↪TRIM(substr(Season,10,25 )) as EndSeason
        From player_reorder
        ;""")

```

[119]: <sqlite3.Cursor at 0x270710d1ac0>

```

[121]: # Reviewing fields in players table
for row in cursor.execute("PRAGMA table_info(player_new)"):
    print(row)

```

```

(0, 'Id', 'TEXT', 0, None, 0)
(1, 'Name', 'TEXT', 0, None, 0)
(2, 'Birthdate', 'TEXT', 0, None, 0)
(3, 'Nationality', 'TEXT', 0, None, 0)
(4, 'Position', 'TEXT', 0, None, 0)
(5, 'Dominant hand', 'TEXT', 0, None, 0)
(6, 'Height (in)', 'REAL', 0, None, 0)
(7, 'Weight (lb)', 'REAL', 0, None, 0)
(8, 'Season', 'TEXT', 0, None, 0)
(9, 'Club', 'TEXT', 0, None, 0)
(10, 'ClubId', 'TEXT', 0, None, 0)
(11, 'LeagueCountry', 'TEXT', 0, None, 0)
(12, 'ActivePlayer', 'TEXT', 0, None, 0)
(13, 'Clubs', 'REAL', 0, None, 0)
(14, 'Teammates', 'REAL', 0, None, 0)
(15, 'Tournaments', 'REAL', 0, None, 0)
(16, 'Matches', 'REAL', 0, None, 0)
(17, 'Awards', 'REAL', 0, None, 0)
(18, 'Ranking', 'REAL', 0, None, 0)
(19, 'Ranking in 2023', 'REAL', 0, None, 0)
(20, 'Player ranking', 'REAL', 0, None, 0)
(21, 'Coach ranking', 'REAL', 0, None, 0)
(22, 'Registered', 'TEXT', 0, None, 0)
(23, 'Views', 'REAL', 0, None, 0)
(24, 'Nickname', 'TEXT', 0, None, 0)
(25, 'Maiden name', 'TEXT', 0, None, 0)
(26, 'Spike (cm)', 'REAL', 0, None, 0)
(27, 'Block (cm)', 'REAL', 0, None, 0)
(28, 'Url', 'TEXT', 0, None, 0)
(29, 'Image', 'TEXT', 0, None, 0)
(30, 'ClubUrl', 'TEXT', 0, None, 0)
(31, 'ClubLogo', 'TEXT', 0, None, 0)
(32, 'LeagueCountryFlag', 'TEXT', 0, None, 0)
(33, 'Agent', 'TEXT', 0, None, 0)
(34, 'Top100', 'TEXT', 0, None, 0)
(35, 'Top200', 'TEXT', 0, None, 0)
(36, 'Top500', 'TEXT', 0, None, 0)

```

```
(37, 'Twitter', 'TEXT', 0, None, 0)
(38, 'Facebook', 'TEXT', 0, None, 0)
(39, 'Instagram', 'TEXT', 0, None, 0)
(40, 'Youtube', 'TEXT', 0, None, 0)
(41, 'Linkedin', 'TEXT', 0, None, 0)
(42, 'Website', 'TEXT', 0, None, 0)
(43, 'Timeline', 'REAL', 0, None, 0)
(44, 'Followers', 'REAL', 0, None, 0)
(45, 'Last activity', 'TEXT', 0, None, 0)
(46, 'Added by', 'TEXT', 0, None, 0)
(47, 'User', 'REAL', 0, None, 0)
(48, 'Recommendations', 'REAL', 0, None, 0)
(49, 'Partners', 'REAL', 0, None, 0)
(50, 'first_name', 'TEXT', 0, None, 0)
(51, 'last_name', 'TEXT', 0, None, 0)
(52, 'middle_name', 'TEXT', 0, None, 0)
(53, 'last_name_final', 'TEXT', 0, None, 0)
(54, 'first_initial', 'TEXT', 0, None, 0)
(55, 'StartSeason', '', 0, None, 0)
(56, 'EndSeason', '', 0, None, 0)
```

```
[123]: # Reviewing fields in team roster table
for row in cursor.execute("PRAGMA table_info(team_rosters)"):
    print(row)
```

```
(0, 'Team', 'TEXT', 0, None, 0)
(1, 'JerseyNumber', 'TEXT', 0, None, 0)
(2, 'Name', 'TEXT', 0, None, 0)
(3, 'Position', 'TEXT', 0, None, 0)
(4, 'TeamName', 'TEXT', 0, None, 0)
```

```
[125]: # Reviewing fields in team results table
for row in cursor.execute("PRAGMA table_info(team_results)"):
    print(row)
```

```
(0, 'idEvent', 'TEXT', 0, None, 0)
(1, 'strHomeTeam', 'TEXT', 0, None, 0)
(2, 'strAwayTeam', 'TEXT', 0, None, 0)
(3, 'intHomeScore', 'INTEGER', 0, None, 0)
(4, 'intAwayScore', 'INTEGER', 0, None, 0)
(5, 'intRound', 'TEXT', 0, None, 0)
(6, 'winning_team', 'TEXT', 0, None, 0)
(7, 'dateEvent', 'TEXT', 0, None, 0)
(8, 'strSport', 'TEXT', 0, None, 0)
(9, 'idLeague', 'TEXT', 0, None, 0)
(10, 'strLeague', 'TEXT', 0, None, 0)
(11, 'strSeason', 'TEXT', 0, None, 0)
(12, 'strTime', 'TEXT', 0, None, 0)
(13, 'idHomeTeam', 'TEXT', 0, None, 0)
```

```
(14, 'idAwayTeam', 'TEXT', 0, None, 0)
```

```
[131]: # Need to summarize team results to summarize their Olympic performance as the
        ↳ table has two teams per row
```

```
cursor.execute("""
    CREATE TABLE team_stats AS

    Select team_name,
    Case when team_name = 'Brazil' then 'BRA'
    when team_name = 'China' then 'CHI'
    when team_name = 'Dominican Republic' then 'DOM'
    when team_name = 'France' then 'FRA'
    when team_name = 'Italy' then 'ITA'
    when team_name = 'Japan' then 'JAP'
    when team_name = 'Netherlands' then 'NET'
    when team_name = 'Poland' then 'POL'
    when team_name = 'Serbia' then 'SER'
    when team_name = 'Turkey' then 'TUR'
    when team_name = 'USA' then 'USA'
    Else NULL End as team_short,
    sum(SetsWon) as TotalSets, sum(GamesWon) as TotalGames
    From (
        SELECT strHomeTeam as team_name, sum(intHomeScore) as SetsWon, Sum(Case
        ↳ When intHomeScore > intAwayScore Then 1 Else 0 End) as GamesWon FROM
        ↳ team_results Group by strHomeTeam
        UNION ALL
        SELECT strAwayTeam as team_name, sum(intAwayScore) as SetsWon, Sum(Case
        ↳ When intAwayScore > intHomeScore Then 1 Else 0 End) as GamesWon FROM
        ↳ team_results Group by strAwayTeam
        ) Group by team_name
    ;""")
```

```
[131]: <sqlite3.Cursor at 0x270710d1ac0>
```

```
[133]: # Reviewing fields in team results table
        for row in cursor.execute("PRAGMA table_info(team_stats)":
            print(row)
```

```
(0, 'team_name', 'TEXT', 0, None, 0)
(1, 'team_short', '', 0, None, 0)
(2, 'TotalSets', '', 0, None, 0)
(3, 'TotalGames', '', 0, None, 0)
```

```
[141]: # Merging team rosters with team stats
        # Also splitting first initial and last name
        cursor.execute("""
            CREATE TABLE team_roster_stats AS
            Select a.Team, a.JerseyNumber, a.Name,
```

```

        Case When a.TeamName in ('Brazil','Dominican
↳Republic','France','Italy','Japan','Kenya','Turkey') Then a.Name
        When a.TeamName in ('Serbia') and a.Name = 'PopoviÄ M.' then 'PopoviÄ'
        When a.TeamName in ('Serbia') and a.Name <> 'PopoviÄ M.' then a.Name
        When a.TeamName in ('Netherlands') and a.Name <> 'M. Jasper' then a.Name
        When a.TeamName in ('Netherlands') and a.Name = 'M. Jasper' then 'Jasper'
        When a.TeamName in ('USA') and a.Name <> 'A. Skinner' then a.Name
        When a.TeamName in ('USA') and a.Name = 'A. Skinner' then 'Skinner'
        When a.TeamName in ('Poland') then TRIM(substr(a.Name,3,40 ))
        When a.TeamName in ('China') and a.Name in ('Yuan X.Y.','Diao L.Y.','Gong X.
↳Y.','Wang Y.Y.','Zhang C.N.','Li Y.Y.',
            'Zheng Y.X.','Wang M.J.','Wu M.J.') then substr(a.Name,-5, -10)
        When a.TeamName in ('China') and a.Name in ('Ding X.','Gao Y.','Zhu T.')
↳then substr(a.Name,-3,-4)
        Else NULL end as LastName,

        Case When a.TeamName in ('USA') and a.Name = 'A. Skinner' then 'A'
        When a.TeamName in ('Netherlands') and a.Name = 'M. Jasper' then 'M'
        When a.TeamName in ('Serbia') and a.Name = 'PopoviÄ M.' then 'M'
        When a.TeamName in ('Poland') then TRIM(substr(a.Name,1,1 ))
        When a.TeamName in ('China') and a.Name in ('Yuan X.Y.','Diao L.Y.','Gong X.
↳Y.','Wang Y.Y.','Zhang C.N.','Li Y.Y.',
            'Zheng Y.X.','Wang M.J.','Wu M.J.') then substr(a.Name,-4, 1)
        When a.TeamName in ('China') and a.Name in ('Ding X.','Gao Y.','Zhu T.')
↳then substr(a.Name,-2,1)
        Else Cast(NULL as varchar(100)) End as FirstInitial,
        a.Position, b.TotalSets, b.TotalGames
    From team_rosters a
    Left Join team_stats b
    on a.Team = b.team_short
;""")

```

[141]: <sqlite3.Cursor at 0x270710d1ac0>

```

[143]: # Reviewing fields in team results table
for row in cursor.execute("PRAGMA table_info(team_roster_stats)":
    print(row)

```

```

(0, 'Team', 'TEXT', 0, None, 0)
(1, 'JerseyNumber', 'TEXT', 0, None, 0)
(2, 'Name', 'TEXT', 0, None, 0)
(3, 'LastName', '', 0, None, 0)
(4, 'FirstInitial', '', 0, None, 0)
(5, 'Position', 'TEXT', 0, None, 0)
(6, 'TotalSets', '', 0, None, 0)
(7, 'TotalGames', '', 0, None, 0)

```

```
[149]: # Merging team rosters stats with players
# and a.FirstInitial = b.first_initial
# and a.Position = b.Position
cursor.execute("""
    CREATE TABLE team_roster_stats_player AS
    Select a.Team, a.JerseyNumber, a.LastName, a.FirstInitial, a.Position, a.
↪TotalSets, a.TotalGames,
    b.Id, b.Name, b.first_name, b.last_name, b.Last_name_final, b.Birthdate, b.
↪Season, b.StartSeason, b.EndSeason,
    b.Nationality, b.Position as PlayerPosition, b.[Height (in)], b.
↪LeagueCountry, b.ActivePlayer,
    b.Tournaments, b.Matches, b.Awards, b.[Spike (cm)], b.[Block (cm)],
    b.Top100, b.Top200, b.Top500
    From team_roster_stats a
    Left Join player_new b
    on a.LastName = b.last_name_final
    and b.EndSeason in ('still','2023/24','2022/23')

;""")
```

```
[149]: <sqlite3.Cursor at 0x270710d1ac0>
```

```
[151]: cursor.execute("""
    DELETE From team_roster_stats_player
    Where Id in ('p100114', 'p100116', 'p100444', 'p100702', 'p101051',
↪'p102512', 'p103083', 'p103736', 'p104676', 'p104685',
'p104723', 'p104797', 'p104868', 'p104984', 'p105009', 'p105112', 'p105307',
↪'p105349', 'p105351', 'p105358', 'p105567', 'p105879',
'p105978', 'p106170', 'p106177', 'p106352', 'p106583', 'p107051', 'p107342',
↪'p107670', 'p107680', 'p107680', 'p107691', 'p107898',
'p10795', 'p109036', 'p109119', 'p110561', 'p110724', 'p11112', 'p11230',
↪'p11230', 'p11357', 'p11358', 'p113681', 'p113681',
'p11401', 'p114828', 'p11552', 'p118007', 'p118216', 'p11921', 'p119897',
↪'p119908', 'p119909', 'p120378', 'p121116', 'p122884',
'p123247', 'p124150', 'p124389', 'p124696', 'p126211', 'p128166', 'p128411',
↪'p128521', 'p132543', 'p136243', 'p13777', 'p138137',
'p139616', 'p142353', 'p143495', 'p144471', 'p144812', 'p1467', 'p147215',
↪'p150358', 'p151949', 'p155023', 'p155964', 'p156212',
'p157640', 'p158859', 'p159420', 'p159961', 'p161387', 'p162681', 'p163539',
↪'p163773', 'p163993', 'p164441', 'p165412', 'p16565',
'p16565', 'p16573', 'p165817', 'p16582', 'p166103', 'p166477', 'p166478',
↪'p167504', 'p168142', 'p169710', 'p170522', 'p170594',
'p171219', 'p172673', 'p175638', 'p177784', 'p178184', 'p18191', 'p21457',
↪'p22457', 'p2424', 'p25233', 'p25543', 'p25543',
'p26121', 'p2648', 'p26717', 'p26717', 'p29137', 'p29139', 'p29582', 'p29650',
↪'p29898', 'p30059', 'p30279', 'p30279', 'p30281',
```

```
'p30283', 'p30288', 'p30288', 'p30291', 'p30294', 'p30295', 'p3087', 'p3087',\
↪ 'p33552', 'p33559', 'p33565', 'p33565', 'p33567',
'p33571', 'p33571', 'p33574', 'p33578', 'p33613', 'p33617', 'p33617', 'p33618',\
↪ 'p33619', 'p33634', 'p33634', 'p33638', 'p33638',
'p33656', 'p3474', 'p3493', 'p3493', 'p3503', 'p40922', 'p40922', 'p40999',\
↪ 'p46025', 'p51903', 'p55116', 'p56154', 'p57690',
'p5801', 'p60123', 'p61986', 'p62548', 'p64426', 'p64494', 'p6609', 'p68588',\
↪ 'p69202', 'p69205', 'p7069', 'p70827', 'p71074',
'p73998', 'p74018', 'p74605', 'p75547', 'p75660', 'p76928', 'p76928', 'p78823',\
↪ 'p79813', 'p79815', 'p79815', 'p79821', 'p79824',
'p79824', 'p79832', 'p79832', 'p79834', 'p79834', 'p79837', 'p79841', 'p79841',\
↪ 'p79852', 'p79865', 'p79867', 'p79867', 'p79879',
'p79886', 'p79886', 'p79887', 'p79890', 'p79945', 'p79956', 'p79956', 'p79979',\
↪ 'p79981', 'p80984', 'p82782', 'p83523', 'p84250',
'p84251', 'p90194', 'p90198', 'p93234', 'p93699', 'p93702', 'p93735', 'p94692',\
↪ 'p94778', 'p94870', 'p95021', 'p95711', 'p95854',
'p96429', 'p96979', 'p97125', 'p97673', 'p98137', 'p98591', 'p99031', 'p99183')
;""")
```

[151]: <sqlite3.Cursor at 0x270710d1ac0>

Based on my data elements, the matching proved quite difficult. Hence the manual review of the results and the deleting of records manually. I'll speak to this issue and potential solutions in my summary.

4.2 Visualizing the Merged Data

```
[155]: # Move tables needed for visuals back to base dataframes for matplotlib
import matplotlib.pyplot as plot
%matplotlib inline

query1 = "SELECT * FROM team_roster_stats_player"
TRSP = pd.read_sql_query(query1, conn)

query2 = "SELECT * FROM team_roster_stats"
TRS = pd.read_sql_query(query2, conn)
```

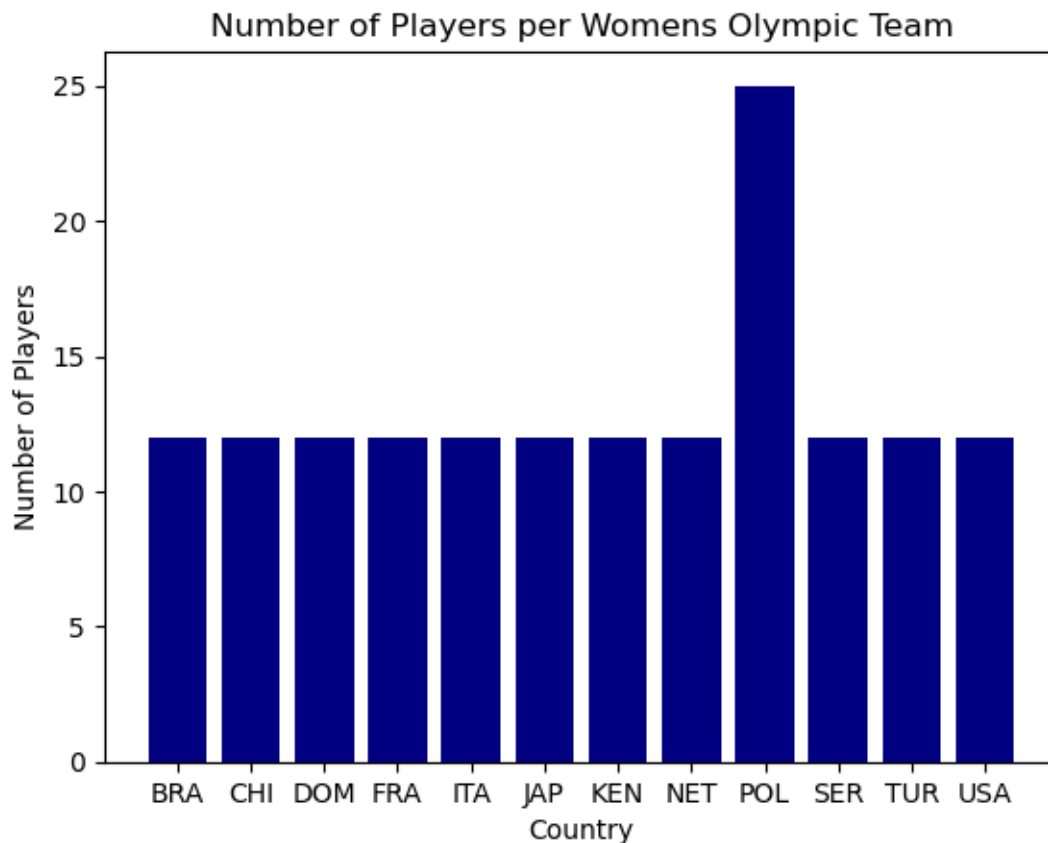
[157]: TRS.head()

```
[157]:   Team JerseyNumber      Name  LastName FirstInitial      Position \
0   USA              2    Poulter    Poulter         None      Setter
1   USA              3  A. Skinner    Skinner          A  Middle-Blocker
2   USA              4 Wong-Orantes Wong-Orantes      None      Libero
3   USA              7   Carlini    Carlini         None      Setter
4   USA             10    Larson    Larson         None  Middle-Blocker

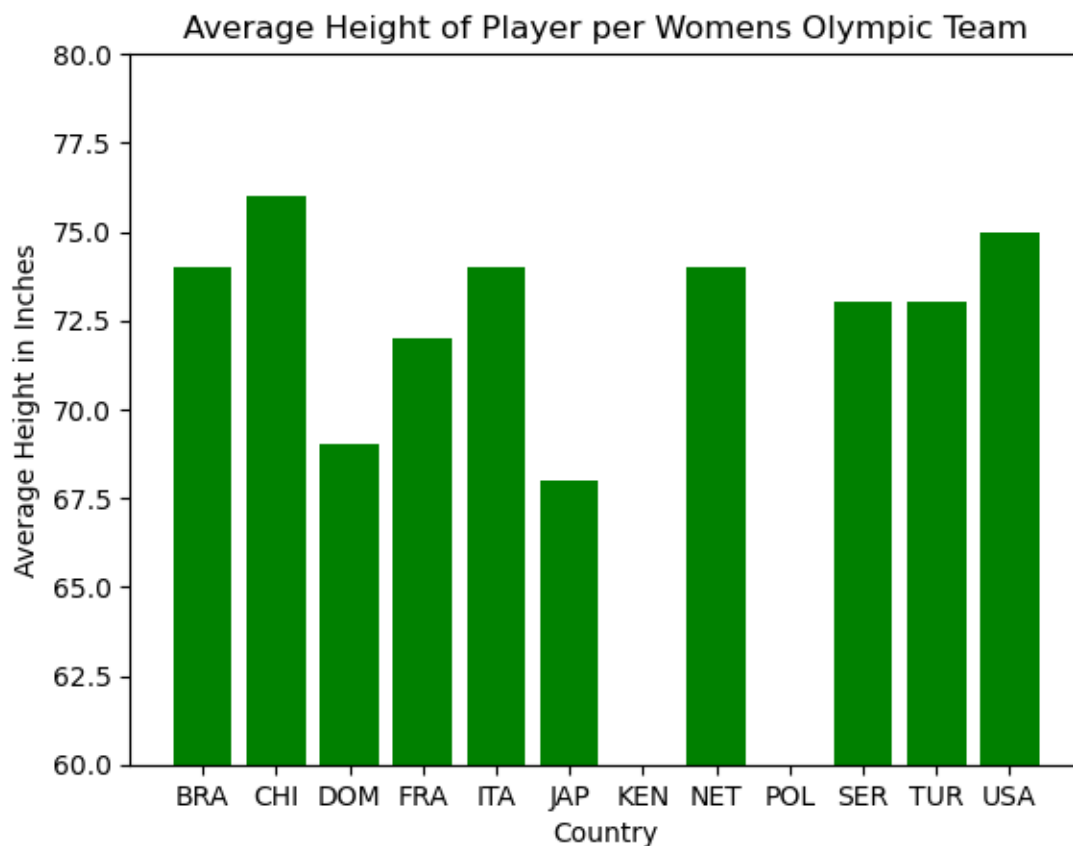
TotalSets  TotalGames
```

0	14.0	4.0
1	14.0	4.0
2	14.0	4.0
3	14.0	4.0
4	14.0	4.0

```
[159]: # First Visual
# Number of Players per Olympic Team
V1 = TRS[['Team', 'JerseyNumber']]
V1 = V1.groupby('Team')
V1 = V1.count()
V1 = V1.reset_index()
# Plot the counts on a bar chart.
plot.bar(x=V1['Team'], height=V1['JerseyNumber'], color='navy')
plot.title("Number of Players per Womens Olympic Team")
plot.xlabel("Country")
plot.ylabel("Number of Players")
plot.show()
```



```
[161]: # Second Visual
# Average Height of Player per Olympic Team
V2 = TRSP[['Team', 'Height (in)']]
V2 = V2.groupby('Team')
V2 = V2['Height (in)'].mean()
V2 = V2.reset_index()
V2['Height (in)'] = V2['Height (in)'].fillna(0)
V2['Height (in)'] = round(V2['Height (in)'])
# Plot the counts on a bar chart.
plot.bar(x=V2['Team'], height=V2['Height (in)'], color='green')
plot.title("Average Height of Player per Womens Olympic Team")
plt.ylim(60, 80)
plot.xlabel("Country")
plot.ylabel("Average Height in Inches")
plot.show()
```

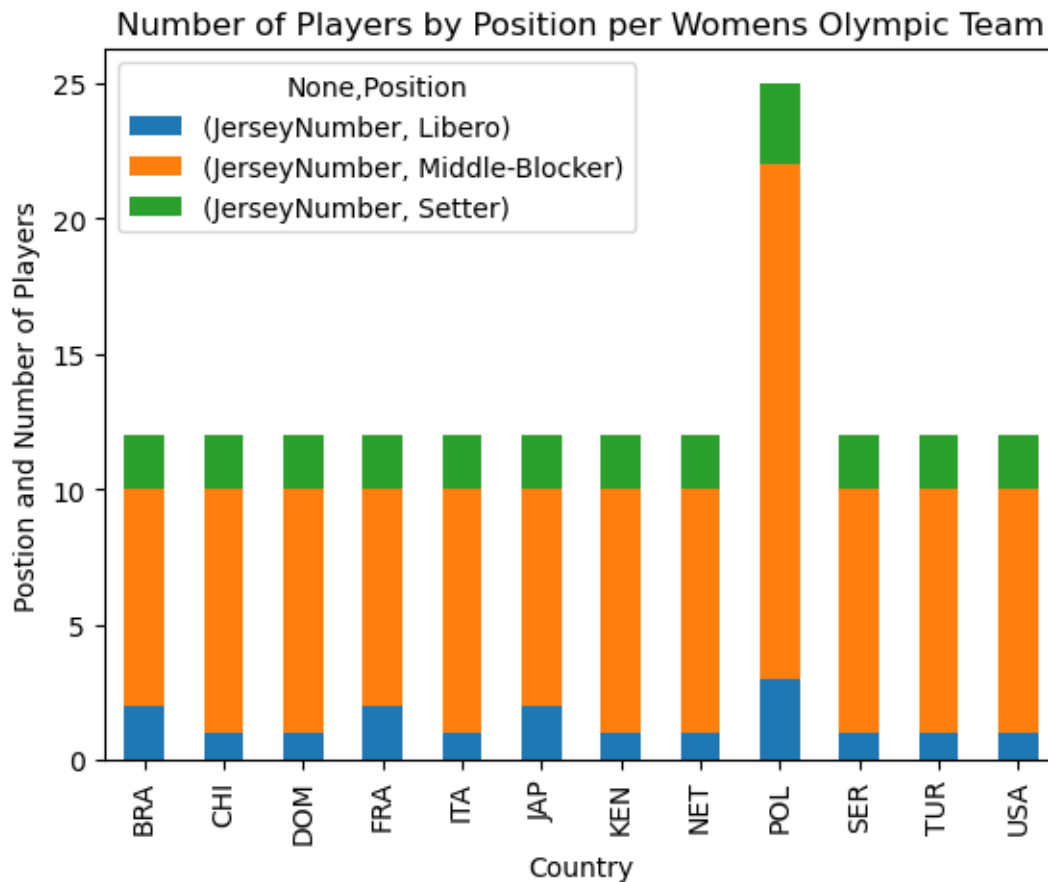


```
[163]: # Third Visual
# Number of Players per Olympic Team by Position
V3 = TRS[['Team', 'Position', 'JerseyNumber']]
V3 = V3.groupby(['Team', 'Position']).count()
```

```

V3 = V3.unstack('Position')
V3Plot = V3.columns.tolist()
# Plot the counts on a bar chart.
V3[V3Plot].plot(kind='bar', stacked=True)
plot.title("Number of Players by Position per Womens Olympic Team")
plot.xlabel("Country")
plot.ylabel("Postion and Number of Players")
plot.show()
# The JerseyNumber field was needed for the count, but I couldn't get the
↳ legend to update to just the position.

```



```

[165]: # Fourth Visual
# Number Games Played Before Olympics by Team
V4 = TRSP[['Team', 'Matches']]
V4['Matches'] = V4['Matches'].fillna(0)
V4 = V4.groupby('Team').sum()
V4 = V4.reset_index()
# Plot the counts on a bar chart.
plot.bar(x=V4['Team'], height=V4['Matches'], color='red')

```

```

plot.title("Number of Gamed Played by Matched Players Before Olympics by Team")
plot.xlabel("Country")
plot.ylabel("Number of Games")
plot.show()

```

C:\Users\caiti\AppData\Local\Temp\ipykernel_12896\2444214743.py:4:

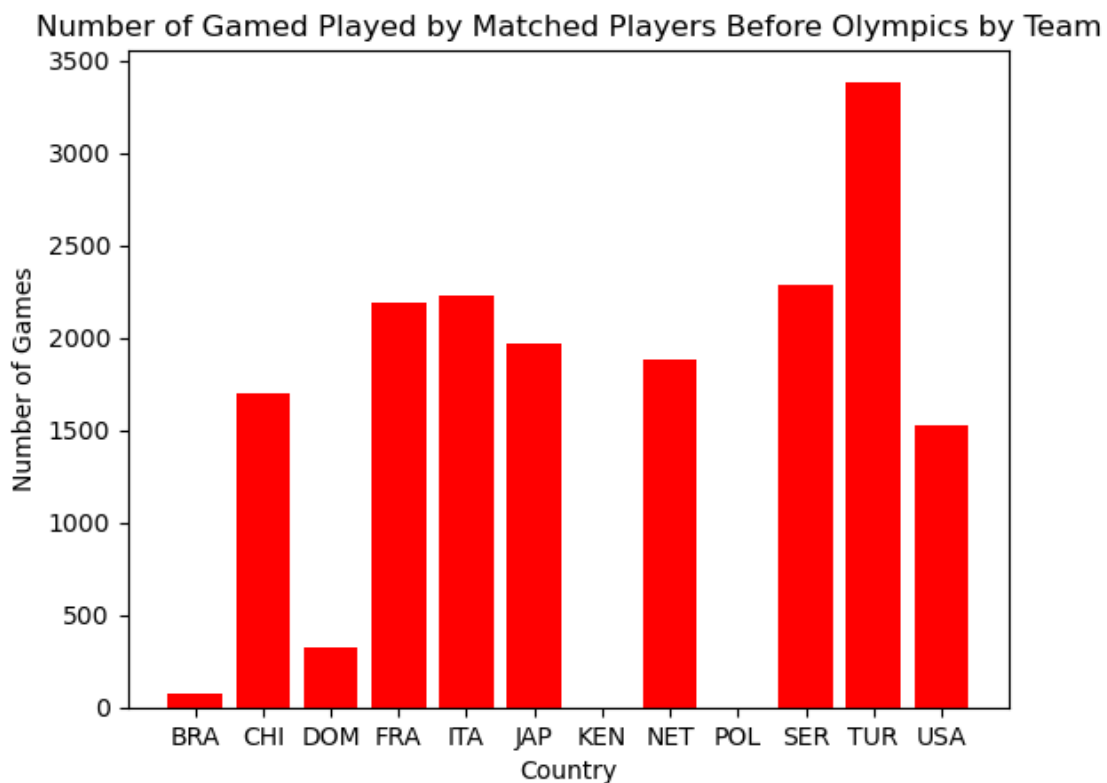
SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.

Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
V4['Matches'] = V4['Matches'].fillna(0)
```

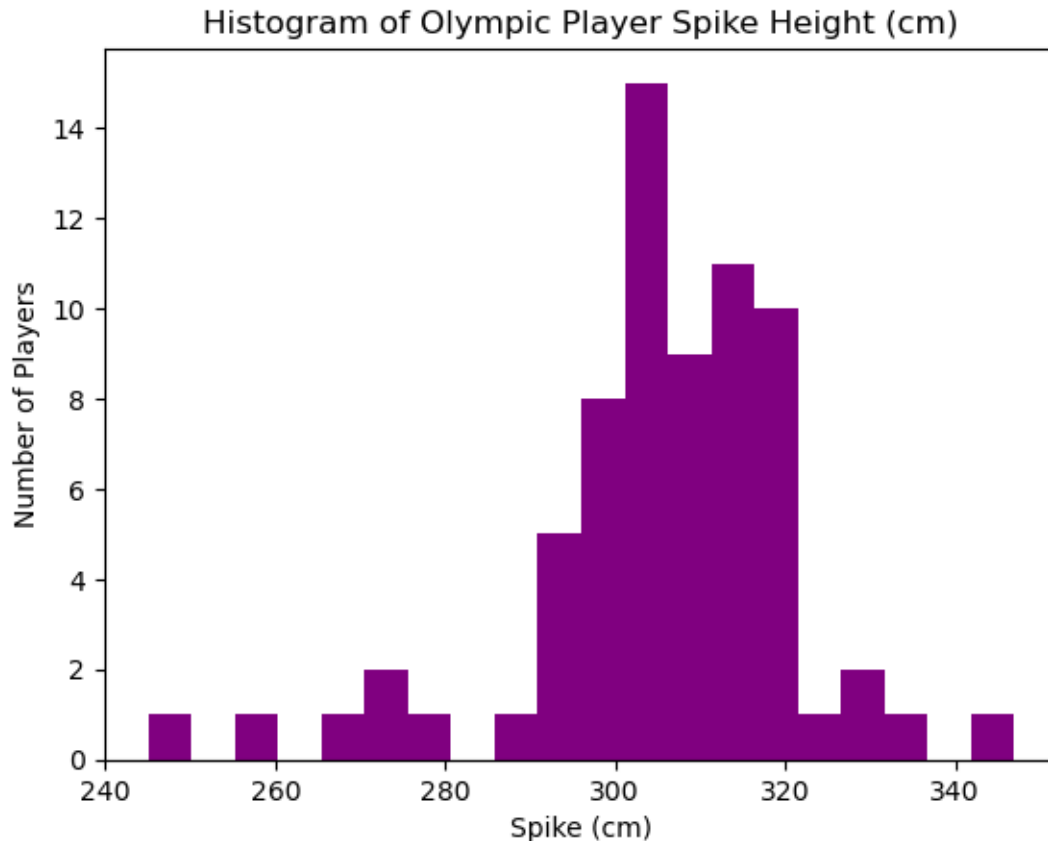


```

[167]: # Fifth Visual
# Histogram of players by spike height
V5 = TRSP[['Spike (cm)']]
V5 = V5.rename(columns={'Spike (cm)': 'Spike'})
V5['Spike'] = V5['Spike'].fillna(0)
V5 = V5.drop(V5[V5['Spike'] == 0].index)
# Plot the values in a histogram
plot.hist(V5.Spike, bins=20, color='purple')

```

```
plot.title('Histogram of Olympic Player Spike Height (cm)')
plot.xlabel('Spike (cm)')
plot.ylabel('Number of Players')
plot.show()
```



DSC 540 Project Summary – Olympic Women’s Volleyball Team and Player Analysis

During the course of this project I learned a lot, including more coding in python, connecting to APIs and websites for data, and the necessity for finding the right data for analysis.

The first data set was a list of a women volleyball players. The issue with this data set was that it was updated by the general public. This caused some duplication and errors within the data itself. Additionally, the structure of the data field (especially name) was not in the ideal number of columns for connecting to other tables. To be able to join I had to write a transformation to attempt to put first and last name in their own fields.

The second data set was retrieved from a website that touted itself as a conduit for driving the commercial success of volleyball. While an accurate source for retrieving team rosters, there was inconsistent formatting in the name field as well. Sometimes they included a first initial (sometimes it at the beginning of the field, sometimes at the end). The limitation of only have a first initial in some cases made joining to the player data very difficult.

The third data set was retrieved via API for the results of the teams during the Olympics. In this

case it was the team name that needed to be modified. The team name (or Country) wasn't in a field by itself. Also, if you wanted to look at whole team performance you had to modify the data set to have a row for each team for a game rather than one. That way you could add up the performance if it was a home versus away team.

After all the transformations and using SQL to merge the modified tables in a database, I wish I would have found better data sets. There were some cases where the data may have been available in a different format (like a website) but I already had a website to pull data from. Wikipedia has a web page for each Olympic team which includes their roster, but I needed an API source. I think this also is true of any data wrangling assignment, even in the real world. You need to pick the best data based on what is in the data set and the capabilities of it to merge with other data.

As far as ethical implications, all of this data was publicly available. Based on my bias to knowing English as my first language, it may have been very difficult for me to match some of the non-USA team player names based on the unique characters used in their language. Other than that, I believe there were no other ethical implications other than those I dictated during the other milestones.